INCH - POUND

MIL-M-28787/419
24 October 1991

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

CONVERTER, A/D MODULE, KEY CODE JFJ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic A/D Converter module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5963

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 3.60849 failures per million operating hours for class I or 11.8760 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be JFJ.

3.6.2 Size. The module size shall be 1A Format A.

3.6.3 Weight. The module shall have a maximum weight of 35 grams.

3.6.4 Contact pins. The module shall have 40 contact pins positioned 1 through 40.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation initial	II	S	10	INF	10	INF	10	INF	MOhm
	end-of-life		S	1	INF	1	INF	1	INF	MOhm
A2-A10	Isolation initial	II	X	10	INF	10	INF	10	INF	MOhm
	end-of-life		X	1	INF	1	INF	1	INF	MOhm
B1-B3	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc stress	IV	S	---------- Module shall pass all remaining tests ----------						
D1-D12	Ioe	V	S	-40	0	-40	0	-40	0	mA
E1-E4	Iih	VI	X	-1	50	-1	50	-1	50	uA
F1-F4	Iil	VII	X	-50	1	-50	1	-50	1	uA
G1-G13	Voh	VIII	X	2.4	5.0	2.4	5.0	2 4	5.0	V
H1-H13	Vol	IX	X	0.0	0.4	0.0	0.4	0.0	0.4	V
J1-J12	Iozh	X	X	-4	40	-4	40	-4	40	uA
K1-K12	Iozl	XI	X	-40	4	-40	4	-40	4	uA
L1	PSFC (5V)	XII	S	1.65	2.75	1.65	2.75	1.65	2.75	uF
L2, L3	PSFC (15V)	XII	S	5 10	8 50	5 10	8.50	5 10	8.50	uF
M1-M4	Cin	XIII	Q	1	20	1	20	1	20	pF
N1	Icc (5 V)	XIV	X	0.1	40.0	0.1	40.0	0.1	40.0	mA
N2	Icc (+15 V)	XIV	X	0.1	7.0	0.1	7.0	0.1	7.0	mA
N3	Icc (-15 V)	XIV	X	-32.0	-0.1	-32 0	-0 1	-32 0	-0.1	mA

- See footnotes at end of table

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 + 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
P1	Unipolar zero (20 V)	XV	X	-0.0122	0.0171	-0.0122	0.0171	-0.0122	0.0171	V
P2	Unipolar zero (10 V)	XV	X	-0.0061	0.0085	-0 0061	0.0085	-0.0061	0.0085	V
P3	Unipolar midscale (10 V)	XV	X	4.9915	5.0061	4 9915	5.0061	4.9915	5.0061	V
P4	Unipolar midscale (20 V)	XV	X	9.9829	10.0122	9 9829	10.0122	9.9829	10.0122	V
P5	Unipolar full scale (10 V)	XV	X	9.9890	10 0037	9 9890	10 0037	9.9890	10.0037	V
P6	Unipolar full scale (20 V)	XV	X	19.9780	20.0073	19.9780	20 0073	19.9780	20.0073	V
Q1	Bipolar zero (±10V)	XVI	X	-0.0171	0.0122	-0 0171	0.0122	-0.0171	0.0122	V
Q2	Bipolar zero (±5 V)	XVI	X	-0.0085	0.0061	-0 0085	0 0061	-0.0085	0.0061	V
Q3	Bipolar neg. full scale (±5V)	XVI	X	-5.0061	-4.9915	-5.0061	-4.9915	-5.0061	-4.9915	V
Q4	Bipolar pos. full scale (±5V)	XVI	X	4.9890	5.0037	4 9890	5 0037	4.9890	5.0037	V
Q5	Bipolar neg. full scale(±10V)	XVI	X	-10.0122	-9.9829	-10 0122	-9 9829	-10.0122	-9.9829	V
Q6	Bipolar pos. full scale(±10V)	XVI	X	9.9780	10.0073	9.9780	10 0073	9.9780	10.0073	V

- See footnotes at end of table

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R1-R4095	Integral linearity	XVII	X	-0.5	0.5	-0.5	0.5	-0.5	0.5	LSB
S1-S4094	Differential linearity	XVIII	X	-0.75	0.75	-0.75	0.75	-0.75	0.75	LSB
T	8-bit conversion	XIX	X	See table XIX						
U1	STS to CE delay	XX	S	0	300	0	300	0	300	nS
U2	Conversion time(8-bit)	XX	S	10	24	10	24	10	24	uS
U3	Conversion time (12-bit)	XX	S	15	35	15	35	15	35	uS
V1	Data out CE low	XXI	S	25	150	25	150	25	150	nS
V2	Access time	XXI	S	0	250	0	250	0	250	nS
W1	Input impedance	XXII	S	6.0	17.0	6.0	17.0	6.0	17.0	KOhms
W2	Input impedance	XXII	S	3.0	8.5	3.0	8.5	3.0	8.5	KOhms
X	Reference voltage, loaded	XXIII	X	9.95	10.05	9.95	10.05	9.95	10.05	V
Y1	PSTR (5V)	XXIV	Q	0	400	0	400	0	400	mVp
Y2, Y3	PSTR (15V)	XXIV	Q	0	700	0	700	0	700	mVp

- See footnotes on next page

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (one percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/419-1 for class I and M28787/419-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram and pin assignments shall conform to figure 1.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for power supply voltages of 5 V dc (Vcc), +15 V dc (Vee), and -15 V dc (-Vee), except that the limits for output short circuit current apply only for power supply voltages of 5.5 V dc (Vcc), +16.5 V dc (Vee), and -16.5 V dc (-Vee).

3.11 Parallel outputs. Paralleling of outputs is not permissible.

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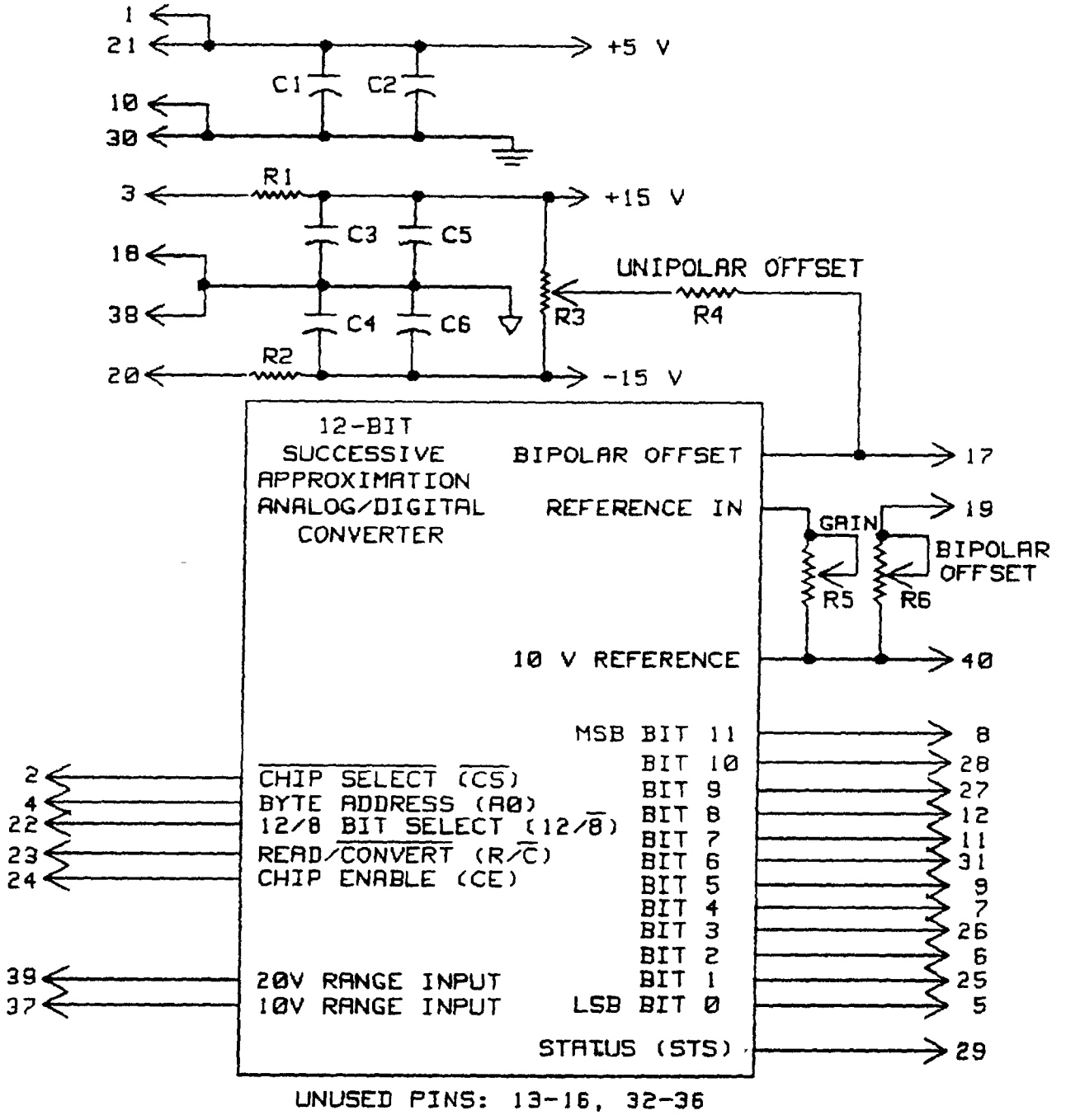


FIGURE 1. JFJ functional block diagram.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through Y shall be performed in any order with the following exceptions. Test conditions A, C and D are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C or D are part of a sequence of tests, the order of testing shall be as follows: A, C, and D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Y in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B and E through Y shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

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4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition L, shall be performed using the initial limits at 25 ± 5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests B and E through Y shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B and E through Y shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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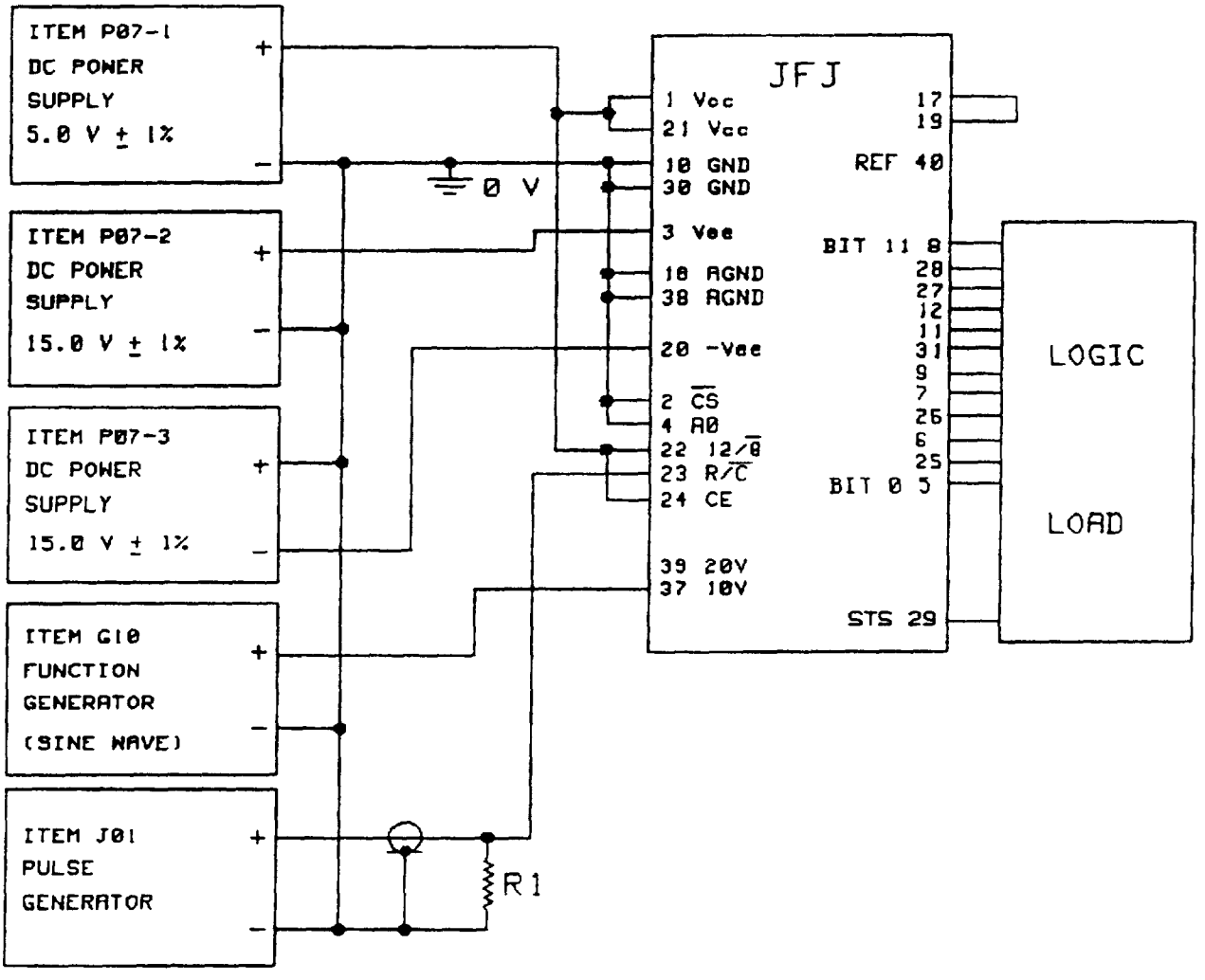


FIGURE 2. Operating temperature, component temperature, and end-of-life test circuit.

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NOTES:

1. R1 = 50 ohms ± 1 percent, 1 W

2. Instrument settings.

   (a) Item J01:
   Inverted pulse
   Vmax = 3.0 ± 0.1 V
   Vmin = 0.0 ± 0.1 V
   PW = .1 ± 0.1 uS
   tTLH = 20 ± 2 nS (10 percent to 90 percent amplitude)
   tTHL = 20 ± 2 nS (10 percent to 90 percent amplitude)
   PRP = 50 ± 5 uS

   (b) Item G10:
   10 V peak to peak
   Freq. = 1 KHz
   DC offset = 0V

3. See figure 3 for waveforms.

4. Load outputs as per figure 11.

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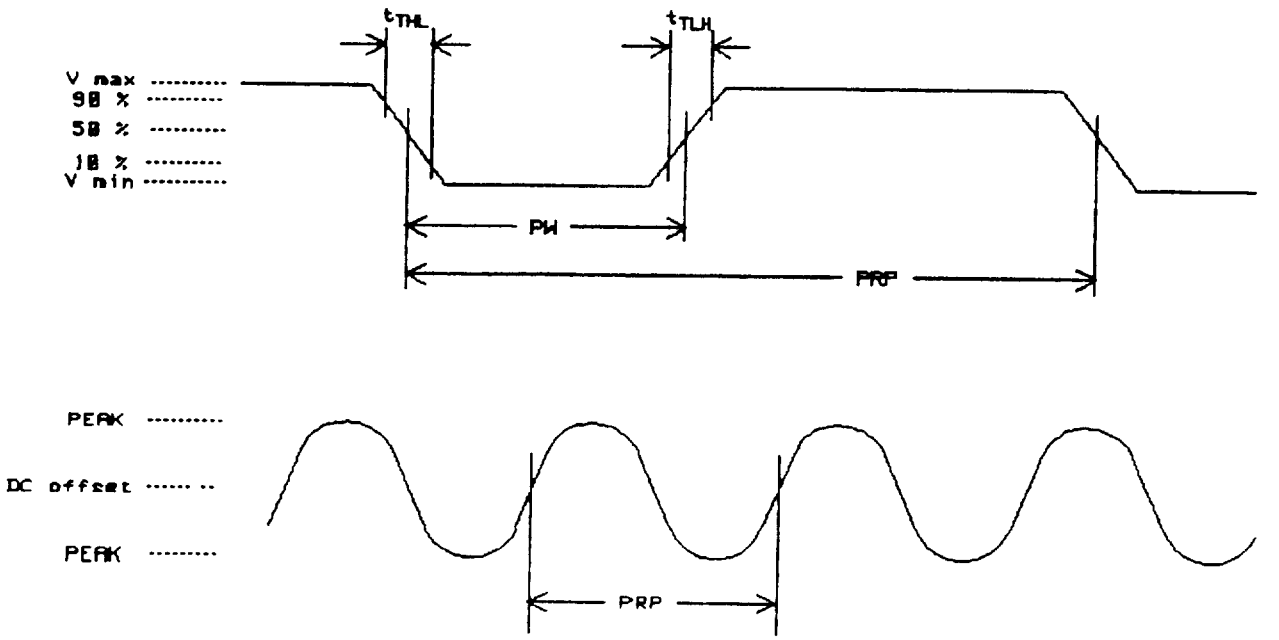


FIGURE 3. Waveforms for instrument set-up.

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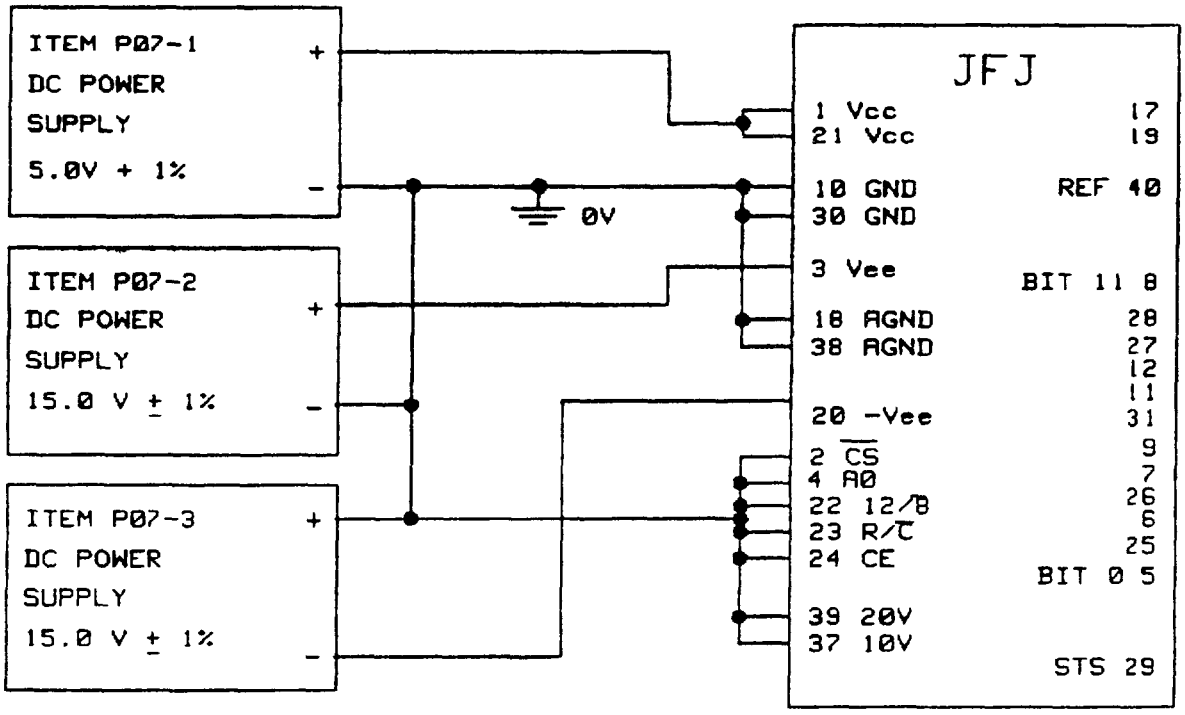


FIGURE 4. Vibration test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ibd = input breakdown current.
(c) Icc = power supply current.
(d) Iih = input high level current.
(e) Iil = input low level current.
(f) Ioh = output high level current.
(g) Iol = output low level current.
(h) Ios = output short circuit current.
(i) Iozh = output high level leakage current.
(j) Iozl = output low level leakage current.
(k) LSB = least significant bit.
(l) MSB = most significant bit.
(m) NA = not applicable.
(n) PSFC = power supply filter capacitance.
(o) PSTA = power supply transient amplitude.
(p) Vih = input high level voltage.
(q) Vil = input low level voltage.
(r) Voh = output high level voltage.
(s) Vol = output low level voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 0.8 W (max); 0.5 W (typical).

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6.3.1 Input parameters.

Signal	Vil max (V)	Vih min (V)	Iil min (uA)	Iih max (uA)	Cin max (pF)
$\overline{CS}$	0.8	2.0	-50	50	20
A0	0.8	2.0	-50	50	20
R/$\overline{C}$	0.8	2.0	-50	50	20
CE	0.8	2.0	-50	50	20
12/$\overline{8}$	NOT TTL COMPATIBLE				

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios min (mA)
BIT 11 (MSB)	2.4	0.4	-0.5	1.6	40	-40	-40
BIT 10	2.4	0.4	-0.5	1.6	40	-40	-40
BIT 9	2.4	0.4	-0.5	1.6	40	-40	-40
BIT 8	2.4	0.4	-0.5	1.6	40	-40	-40
BIT 7	2.4	0.4	-0.5	1.6	40	-40	-40
BIT 6	2.4	0.4	-0.5	1.6	40	-40	-40
BIT 5	2.4	0.4	-0.5	1.6	40	-40	-40
BIT 4	2.4	0.4	-0.5	1.6	40	-40	-40
BIT 3	2.4	0.4	-0.5	1.6	40	-40	-40
BIT 2	2.4	0.4	-0.5	1.6	40	-40	-40
BIT 1	2.4	0.4	-0.5	1.6	40	-40	-40
BIT 0 (LSB)	2.4	0.4	-0.5	1.6	40	-40	-40
STATUS (STS)	2.4	0.4	-0.5	1.6	---	---	---

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6.4 Module function. The JFJ is an 8- or 12-bit analog to digital converter which can be operated in two modes: bipolar, with ranges of -5 to +5 volts and -10 to +10 volts; or unipolar, with ranges of 0 to +10 volts and 0 to +20 volts. When operated as a 12-bit converter, each output bit carries significant data which are read by asserting the read/convert line. As an 8-bit converter, the byte address line controls whether the 8 most significant bits or the 4 least significant bits are read. This 8-bit function allows the JFJ to operate on 8-bit busses. The JFJ also contains a 10 volt reference output.

6.4.1 Functional description of I/O pins.

Functional designation	Pin number	I/O	Functional description
$\overline{CS}$	2	I	Active low chip select.
A0	4	I	Byte address - when low 8 MSBs are enabled. 4 LSBs and 4 trailing zeros are enabled when high.
R/$\overline{C}$	23	I	Data Read or Convert.
CE	24	I	Active high Chip Enable.
12/$\overline{8}$	22	I	12 bit or 8 bit conversion select - must be wired to Vcc or GND.
BIT 11 (MSB)	8	O	Output data bits.
BIT 10	28	O	
BIT 9	27	O	
BIT 8	12	O	
BIT 7	11	O	
BIT 6	31	O	
BIT 5	9	O	
BIT 4	7	O	Output data bits.

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6.4.1 Functional description of I/O pins - Continued.

Functional designation	Pin number	I/O	Functional description
BIT 3	26	O	Output data bits.
BIT 2	6	O	
BIT 1	25	O	
BIT 0 (LSB)	5	O	Output data bits.
STATUS (STS)	29	O	Conversion Status - high while conversion is in progress.
OFFSET	17	O	Connected to pin 19 for bipolar operation, open for unipolar operation.
BIPOLAR	19	O	Connected to pin 17 for bipolar operation, open for unipolar operation.
10 V REF	40	O	10 volt reference output.
10 V INPUT	37	I	Analog input for 10 volt full-scale range.
20 V INPUT	39	I	Analog input for 20 volt full-scale range.

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0265)

Agent: NW

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TABLE II. Isolation test.

Test cond.	Parameter	Connect A to	Connect B to	25 ± 5°C Initial limit Min	25 ± 5°C Initial limit Max	Units	Notes
A1	Frame	All pins	Frame	10	INF	MOhm	1-7
A2	Used to Unused	1-12,17-31, 37-40	13-16, 32-36				
A3	Unused to Unused	13	14-16, 32-36				
A4		14	15,16, 32-36				
A5		15	16,32-36				
A6		16	32-36				
A7		32	33-36				
A8		33	34-36				
A9		34	35,36				
A10	Unused to Unused	35	36	10	INF	MOhm	1-7

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NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V 30 V	± 10 percent ± 10 percent	A1 A2-A10
Isolation measurement	NA	± 500 KOhm	A1-A10
Isolation measurement (See note 6)	NA	± 50 KOhm	A1-A10

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, and 'B' shall correspond to the negative terminal.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. The minimum end-of-life limit is 1 MOhm.

7. Abbreviations used are defined in 6.1.

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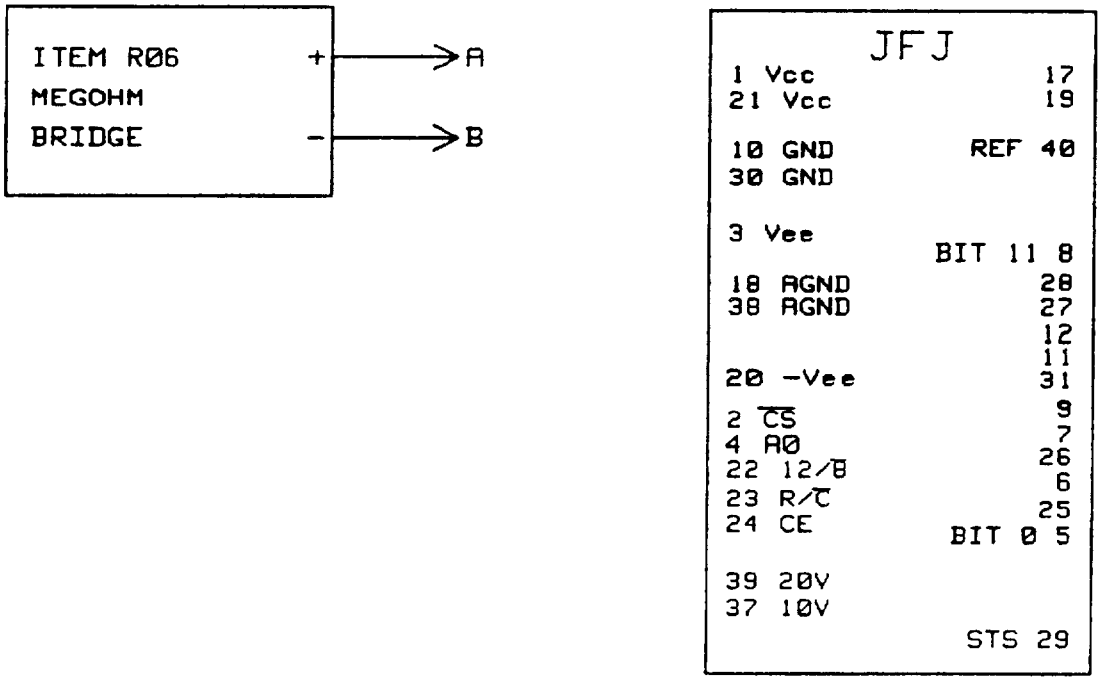


FIGURE 5. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond.	Test description	Set-up conditions Connect A to	Set-up conditions Connect B to	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	5 V	1	21	0	1	Ohm	1-4
B2	GND	10	30	|	|	|	|
B3	AGND	18	38	0	1	Ohm	1-4

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B3

2. The ohmmeter used shall be item R10.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. Abbreviations used are defined in 6.1.

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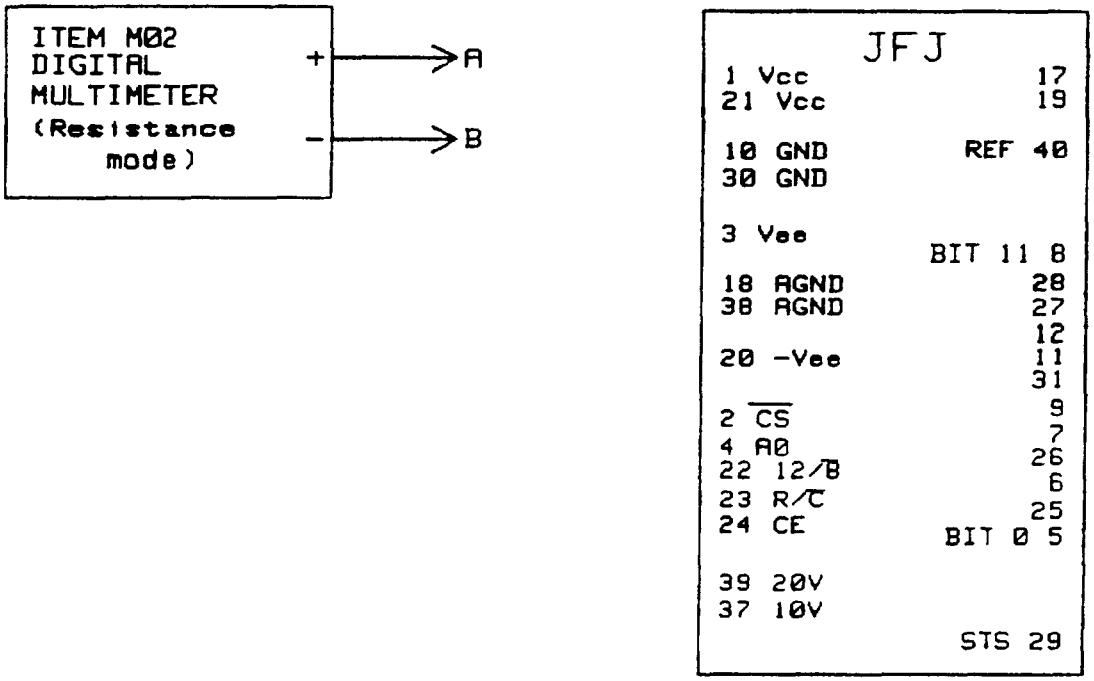


FIGURE 6. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond.	Test description	Set-up conditions Connect A to	Connect B to	25 ± 5°C Initial limits	Notes
C	Power supply stress test	NA	NA	Module shall pass all remaining tests.	1-4

NOTES:

1. Power supply stress test tolerance table.

Condition		Value	Accuracy	Test condition
Power supply,	(Vcc)	7.0 V	+0 mV, -70 mV	C
	(Vee)	16.5 V	+0 mV, -165 mV	C
	(-Vee)	-16.5 V	+165 mV, -0 mV	C
Power supply current limit	(Vcc)	72 mA	NA	C
	(Vee)	12.6 mA	NA	C
	(-Vee)	-57.6 mA	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

4. Abbreviations used are defined in 6.1.

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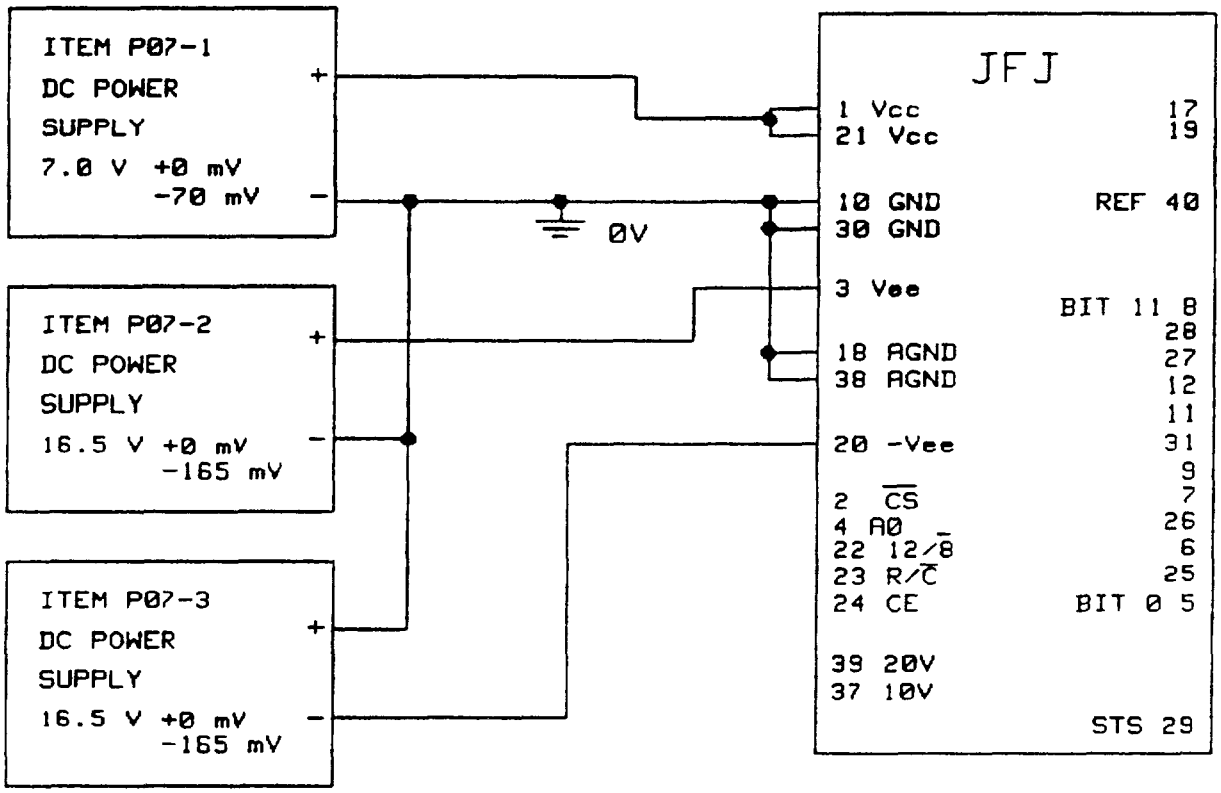


FIGURE 7. Power supply stress test circuit for table IV.

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TABLE V. Output short circuit current test.

Test cond.	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
D1	BIT 11 (MSB)	8	0.0	-40	0	mA	1-4
D2	BIT 10	28					
D3	BIT 9	27					
D4	BIT 8	12					
D5	BIT 7	11					
D6	BIT 6	31					
D7	BIT 5	9					
D8	BIT 4	7					
D9	BIT 3	26					
D10	BIT 2	6					
D11	BIT 1	25					
D12	BIT 0 (LSB)	5	0.0	-40	0	mA	1-4

NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D12
(Vee)	16.5 V	+ 0 mV,-165 mV	D1-D12
(-Vee)	-16.5 V	+ 165 mV,-0 mV	D1-D12

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Power supply current limit (Vcc)	60 mA	NA	D1-D12
(Vee)	10.5 mA	NA	D1-D12
(-Vee)	-48 mA	NA	D1-D12
Forcing voltage, (Vout)	0.0 V	± 20 mV	D1-D12
Pattern voltage high, (VH)	3.0 V	± 100 mV	D1-D12
low, (VL)	0.0 V	± 100 mV	D1-D12
Output current measurement	NA	± 2 mA	D1-D12

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. 'A' shall correspond to the positive terminal and 'B' shall correspond to the negative terminal of the power supply. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Before performing test condition D1, trigger item J01. Item J01 shall not require triggering thereafter.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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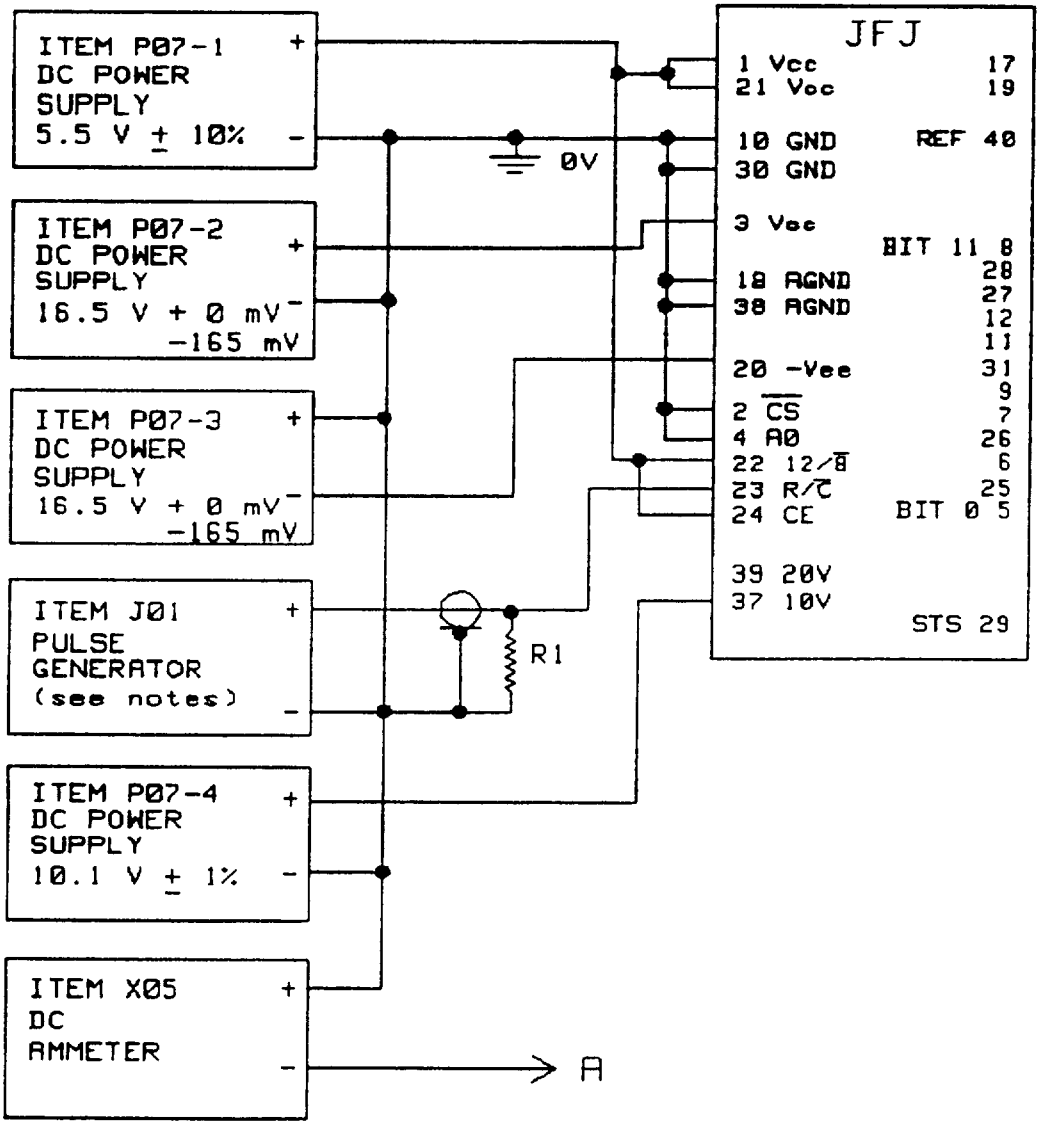


FIGURE 8. Output short circuit current test circuit for table V.

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NOTES:

1. R1 = 50 Ohms ± 1 percent, 1 W

2. Item J01 settings:
   Inverted pulse
   Vmax = 3.0 ± 0.1 V
   Vmin = 0.0 ± 0.1 V
   PW = 1 ± 0.1 uS
   tTLH = 20 ± 2 nS
   tTHL = 20 ± 2 nS
   Single shot

3. See figure 3 for waveforms.

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TABLE VI. Input current, high level, test.

Test cond.	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
E1	$\overline{CS}$	2	2.0	-1	50	uA	1-3
E2	A0	4					
E3	R/$\overline{C}$	23					
E4	CE	24	2.0	-1	50	uA	1-3

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E4
(Vee)	16.5 V	+ 0 mV,-165 mV	E1-E4
(-Vee)	-16.5 V	+ 165 mV,-0 mV	E1-E4
Power supply current limit (Vcc)	60 mA	NA	E1-E4
(Vee)	10.5 mA	NA	E1-E4
(-Vee)	-48 mA	NA	E1-E4
Forcing voltage, (Vih)	2.0 V	± 100 mV	E1-E4
Pattern voltage high, (VH)	3.0 V	± 100 mV	E1-E4
low, (VL)	0.0 V	± 100 mV	E1-E4
Input current measurement	NA	± 2.55 uA	E1-E4

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. 'A' shall correspond to the positive terminal and 'B' shall correspond to the negative terminal of the power supply. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, low level, test.

Test cond.	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	$\overline{CS}$	2	0.8	-50	1	uA	1-3
F2	A0	4					
F3	R/$\overline{C}$	23					
F4	CE	24	0.8	-50	1	uA	1-3

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F4
(Vee)	16.5 V	+ 0 mV,-165 mV	F1-F4
(-Vee)	-16.5 V	+ 165 mV, -0 mV	F1-F4
Power supply current limit (Vcc)	60 mA	NA	F1-F4
(Vee)	10.5 mA	NA	F1-F4
(-Vee)	-48 mA	NA	F1-F4
Forcing voltage, (Vil)	0.8 V	± 40 mV	F1-F4
Pattern voltage high, (VH)	3.0 V	± 100 mV	F1-F4
low, (VL)	0.0 V	± 100 mV	F1-F4
Input current measurement	NA	± 2.55 uA	F1-F4

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NOTES: - Continued.

2. The following procedure is required to perform this test:

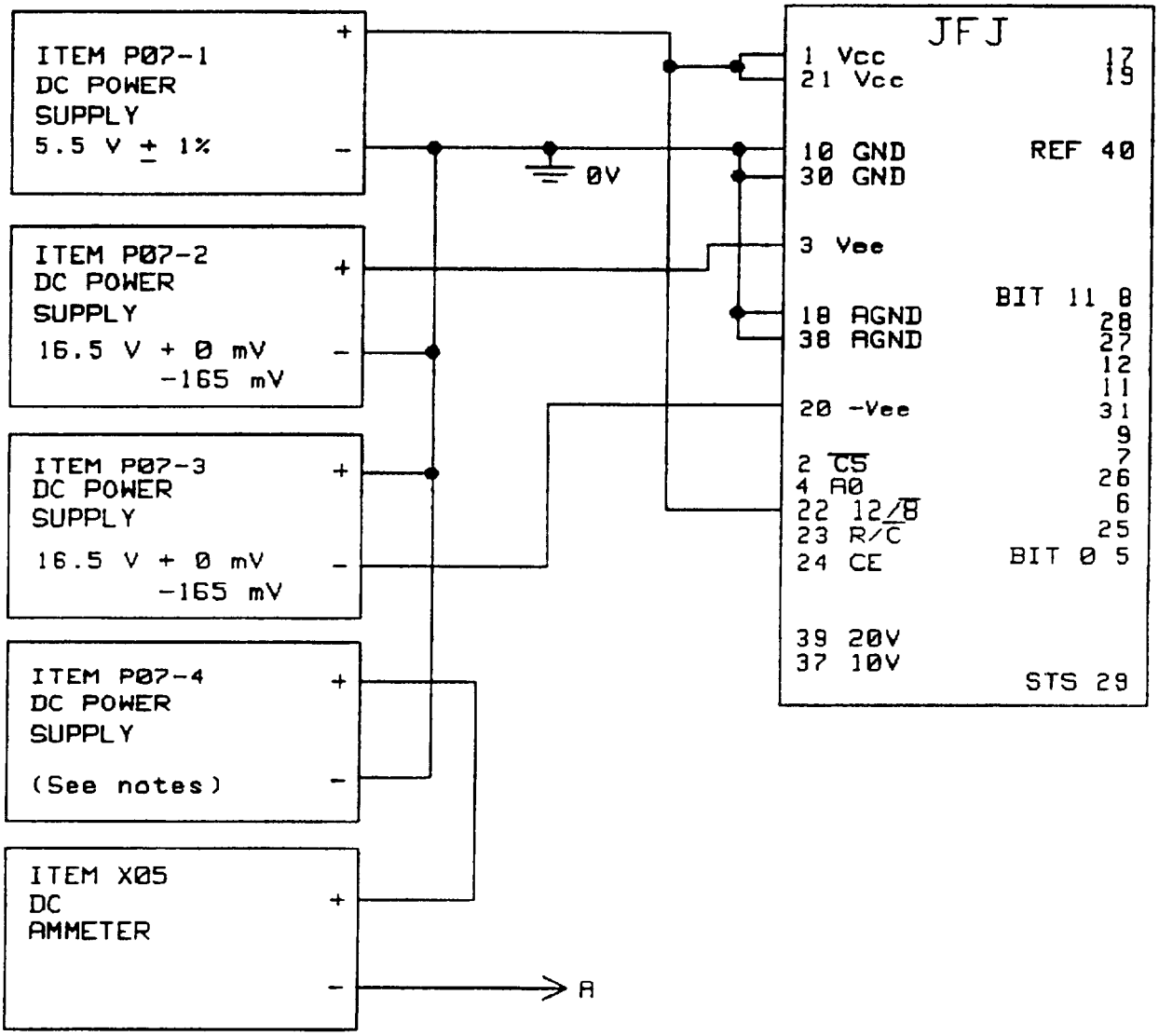
(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. 'A' shall correspond to the positive terminal and 'B' shall correspond to the negative terminal of the power supply. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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NOTE:

1. Item P07-4 set-up:

   2.0 V ± 5 percent for E1 through E4.
   0.8 V ± 5 percent for F1 through F4.

FIGURE 9. Input current test circuit for tables VI and VII.

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TABLE VIII. Output voltage, high level, test.

Test cond.	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G1	BIT 11 (MSB)	8	0.5	2.4	5.0	V	1-6
G2	BIT 10	28					
G3	BIT 9	27					
G4	BIT 8	12					
G5	BIT 7	11					
G6	BIT 6	31					
G7	BIT 5	9					
G8	BIT 4	7					
G9	BIT 3	26					
G10	BIT 2	6					
G11	BIT 1	25					
G12	BIT 0 (LSB)	5					
G13	STATUS (STS)	29	0.5	2.4	5.0	V	1-6

NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	G1-G13
(Vee)	13.5 V	± 135 mV	G1-G13
(-Vee)	-13.5 V	± 135 mV	G1-G13

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Power supply current limit (Vcc)	60 mA	NA	G1-G13
(Vee)	10.5 mA	NA	G1-G13
(-Vee)	-48 mA	NA	G1-G13
Forcing current, (Ioh)	0.5 mA	± 25 uA	G1-G13
Pattern voltage high, (VH)	2.0 V	± 20 mV	G1-G13
low, (VL)	0.8 V	± 8 mV	G1-G13
Output voltage measurement	NA	± 130 mV	G1-G13

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. 'A' shall correspond to the positive terminal and 'B' shall correspond to the negative terminal of the power supply. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Before performing test condition G1, trigger item J01. Item J01 shall not require triggering thereafter.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table VIII and the voltage then measured.

3. During this test, the high and low level input threshold voltage of every input must be verified.

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NOTES: - Continued.

4. For test condition G13, item D05 shall be used for voltage measurement. Item M02 shall be used for G1 through G12.

5. The load shall be as indicated on figure 11.

6. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, low level, test.

Test cond.	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	BIT 11 (MSB)	8	-1.6	0.0	0.4	V	1-6
H2	BIT 10	28					
H3	BIT 9	27					
H4	BIT 8	12					
H5	BIT 7	11					
H6	BIT 6	31					
H7	BIT 5	9					
H8	BIT 4	7					
H9	BIT 3	26					
H10	BIT 2	6					
H11	BIT 1	25					
H12	BIT 0 (LSB)	5					
H13	STATUS (STS)	29	-1.6	0.0	0.4	V	1-6

NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H13
(Vee)	13.5 V	± 135 mV	H1-H13
(-Vee)	-13.5 V	± 135 mV	H1-H13

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Power supply current (Vcc) (Vee) (-Vee)	 60 mA 10.5 mA -48 mA	 NA NA NA	 H1-H13 H1-H13 H1-H13
Forcing current, (Iol)	-1.6 mA	± 80 uA	H1-H13
Pattern voltage high, (VH) low, (VL)	 2.0 V 0.8 V	 ± 20 mV ± 8 mV	 H1-H13 H1-H13
Output voltage measurement	NA	± 20 mV	H1-H13

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. 'A' shall correspond to the positive terminal and 'B' shall correspond to the negative terminal of the power supply. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Before performing test condition H1, trigger item J01. Item J01 shall not require triggering thereafter.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. During this test, the high and low level input threshold voltage of every input must be verified.

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NOTES: - Continued.

4. For test condition H13, item D05 shall be used for voltage measurement. Item M02 shall be used for H1 through H12.

5. The loads shall be as indicated on figure 11.

6. Abbreviations used are defined in 6.1.

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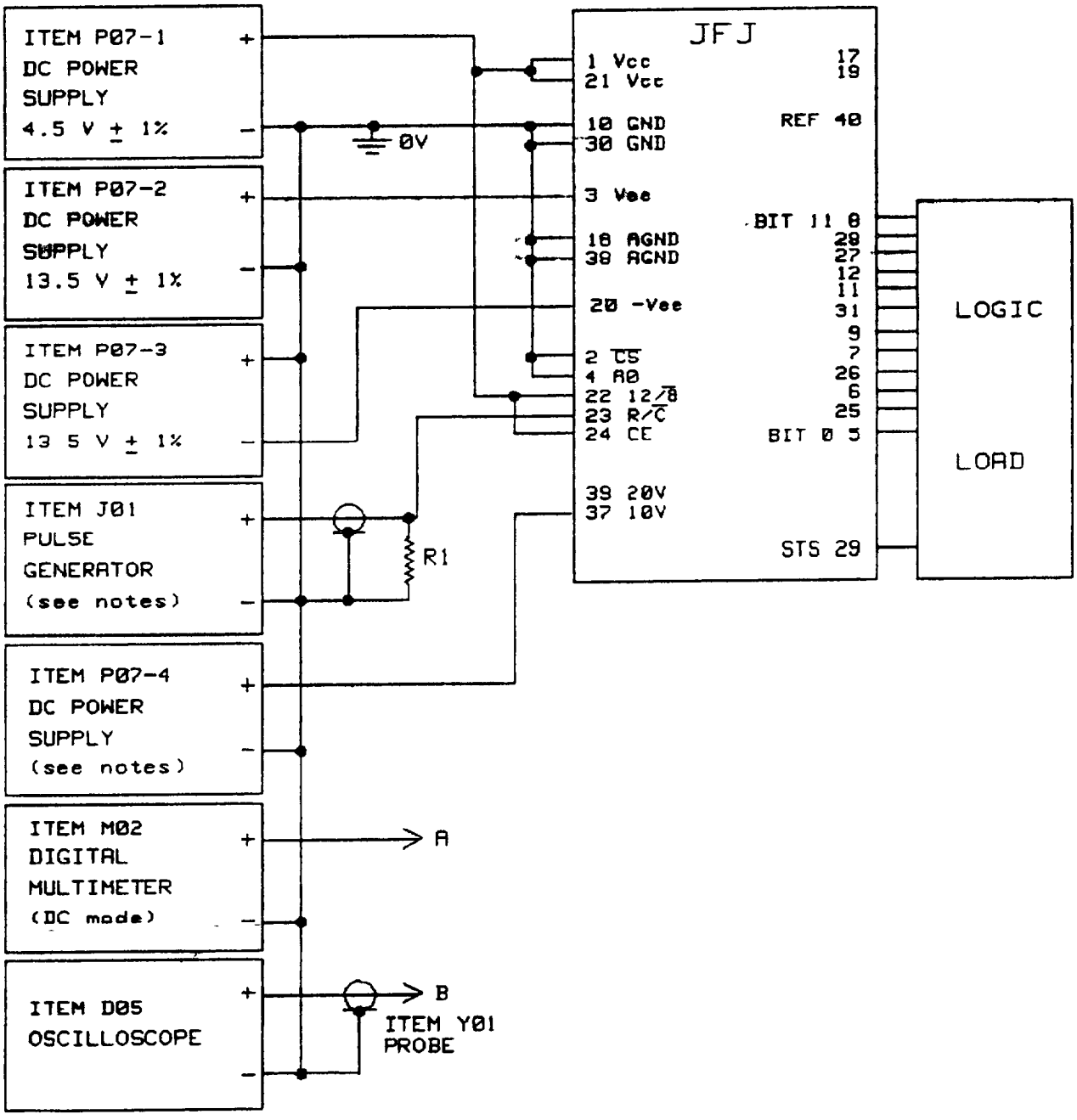


FIGURE 10. Output voltage test circuit for tables VIII and IX.

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NOTES:

1. R1 = 50 Ohms ± 1 percent, 1 W

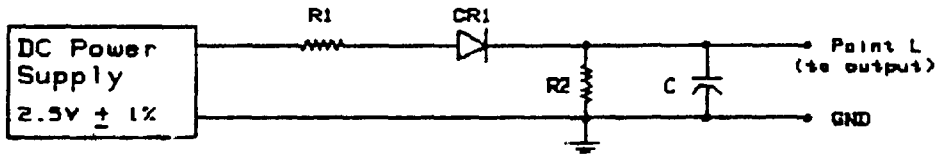
2. Instrument settings:

   (a) Item J01:
   Inverted pulse
   Vmax = 3.0 ± 0.1 V
   Vmin = 0.0 ± 0.1 V
   PW = 1 ± 0.1 uS
   tTLH = 20 ± 2 nS
   tTHL = 20 ± 2 nS
   G1-G12 and H1-H12:
   Single shot
   G13 and H13:
   PRP = 50 ± 5 uS

   (b) Item P07-4:
   10.1 ± 0.1 V (G1 through G13)
   0.0 ± 0.1 V (H1 through H13)

3. See figure 3 for waveforms.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

   (a) C = 150 ± 15 pF (including probe and parasitic capacitance) for sample dynamic tests.

   (b) The functional test shall be performed with only a 150 pF ± 15 pF load (including probe and parasitic capacitance) attached to each output.

   (c) C not required for any other tests.

3. The resistance values are as follows:

   (a) R1 = 845 Ohms

   (b) R2 = 4800 Ohms

FIGURE 11. Logic load circuit.

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TABLE X. Output leakage current, high level, test.

Test cond.	Signal name	Pin no.	Voh (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	BIT 11 (MSB)	8	2.4	-4	40	uA	1-3
J2	BIT 10	28					
J3	BIT 9	27					
J4	BIT 8	12					
J5	BIT 7	11					
J6	BIT 6	31					
J7	BIT 5	9					
J8	BIT 4	7					
J9	BIT 3	26					
J10	BIT 2	6					
J11	BIT 1	25					
J12	BIT 0 (LSB)	5	2.4	-4	40	uA	1-3

NOTES:

1. Output leakage current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	J1-J12
(Vee)	16.5 V	+ 0 mV,-165 mV	J1-J12
(-Vee)	-16.5 V	+ 165 mV,-0 mV	J1-J12

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Power supply current limit (Vcc)	60 mA	NA	J1-J12
(Vee)	10.5 mA	NA	J1-J12
(-Vee)	-48 mA	NA	J1-J12
Forcing voltage, (Voh)	2.4 V	± 120 mV	J1-J12
Pattern voltage high, (VH)	3.0 V	± 100 mV	J1-J12
low, (VL)	0.0 V	± 100 mV	J1-J12
Output current measurement	NA	± 2.20 uA	J1-J12

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. 'A' shall correspond to the positive terminal and 'B' shall correspond to the negative terminal of the power supply. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Before performing test condition J1, trigger item J01. Item J01 shall not require triggering thereafter.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table X and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, low level, test.

Test cond.	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
K1	BIT 11 (MSB)	8	0.4	-40	4	uA	1-3
K2	BIT 10	28					
K3	BIT 9	27					
K4	BIT 8	12					
K5	BIT 7	11					
K6	BIT 6	31					
K7	BIT 5	9					
K8	BIT 4	7					
K9	BIT 3	26					
K10	BIT 2	6					
K11	BIT 1	25					
K12	BIT 0 (LSB)	5	0.4	-40	4	uA	1-3

NOTES:

1. Output leakage current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K1-K12
(Vee)	16.5 V	+ 0 mV,-165 mV	K1-K12
(-Vee)	-16.5 V	+ 165 mV,-0 mV	K1-K12

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Power supply current limit (Vcc)	60 mA	NA	K1-K12
(Vee)	10.5 mA	NA	K1-K12
(-Vee)	-48 mA	NA	K1-K12
Forcing voltage, (Vol)	0.4 V	± 20 mV	K1-K12
Pattern voltage high, (VH)	3.0 V	± 100 mV	K1-K12
low, (VL)	0.0 V	± 100 mV	K1-K12
Output current measurement	NA	± 2.20 uA	K1-K12

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. 'A' shall correspond to the positive terminal and 'B' shall correspond to the negative terminal of the power supply. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Before performing test condition K1, trigger item J01. Item J01 shall not require triggering thereafter.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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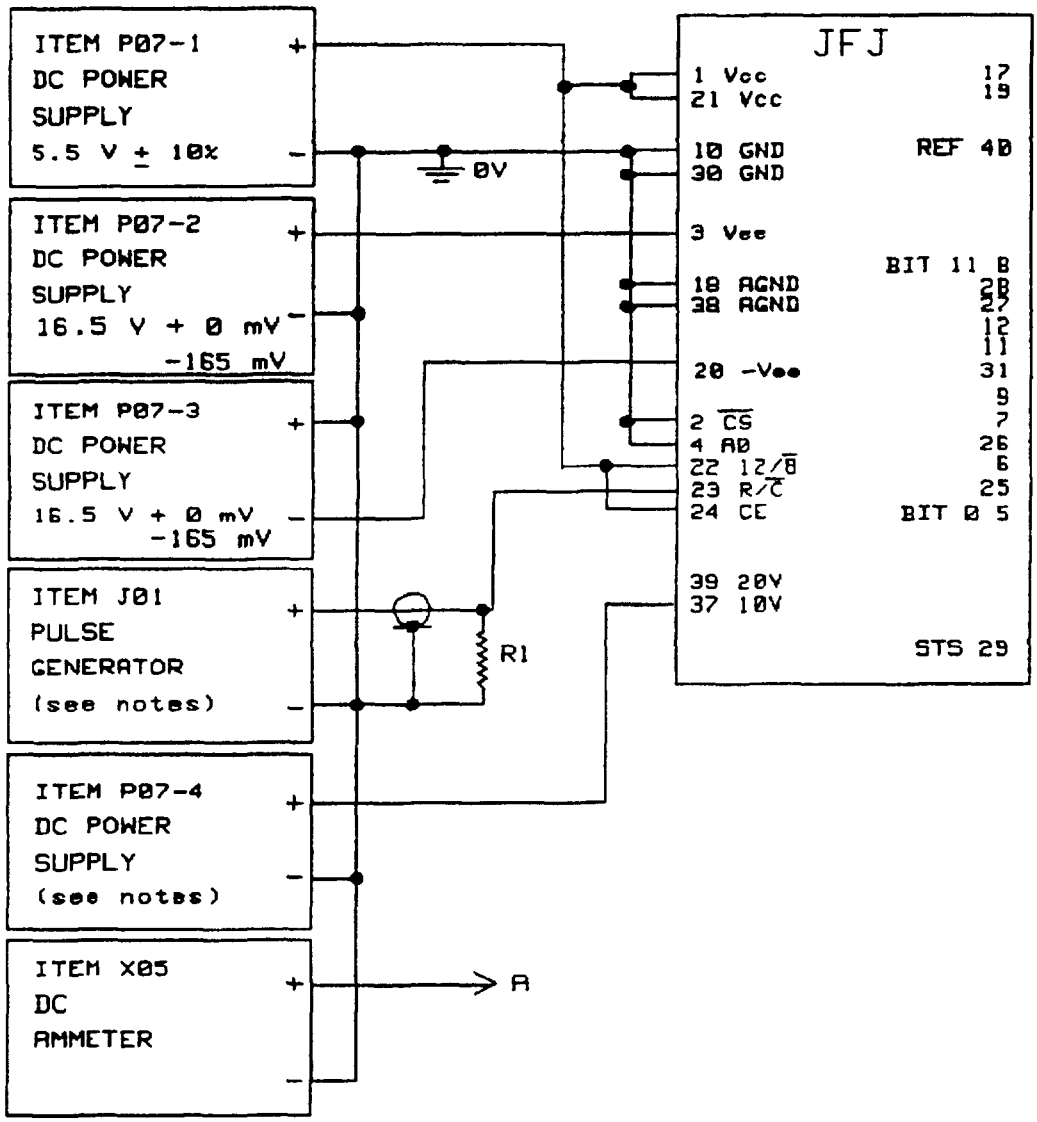


FIGURE 12. Output leakage current, high and low level, test circuit for tables X and XI.

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NOTES:

1. R1 = 50 Ohms ± 1 percent, 1 W

2. Instrument settings:

   (a) Item J01:
   Vmax = 3.0 ± 0.1 V
   Vmin = 0.0 ± 0.1 V
   PW = 1 ± 0.1 uS
   tTLH = 20 ± 2 nS
   tTHL = 20 ± 2 nS
   Single shot

   (b) Item P07-4 set-up:
   10.1 ± 0.1 V (J1 through J12)
   0.0 ± 0.1 V (K1 through K12)

3. See figure 3 for waveforms.

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TABLE XII. Power supply filter capacitance test.

Test cond.	Signal name	Set-up Conditions		25 ± 5°C Initial limits		Units	Notes
		Connect A to	Connect B to	Min	Max		
L1	PSFC	1	10	1.65	2.75	uF	1-4
L2	PSFC	3	18	5.10	8.50	uF	1-4
L3	PSFC	18	20	5.10	8.50	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	L1-L3
Capacitance bridge voltage	0.3 Vp	± 30 mVp	L1-L3
Capacitance measurement	NA NA	± 55 nF ± 170 nF	L1 L2, L3

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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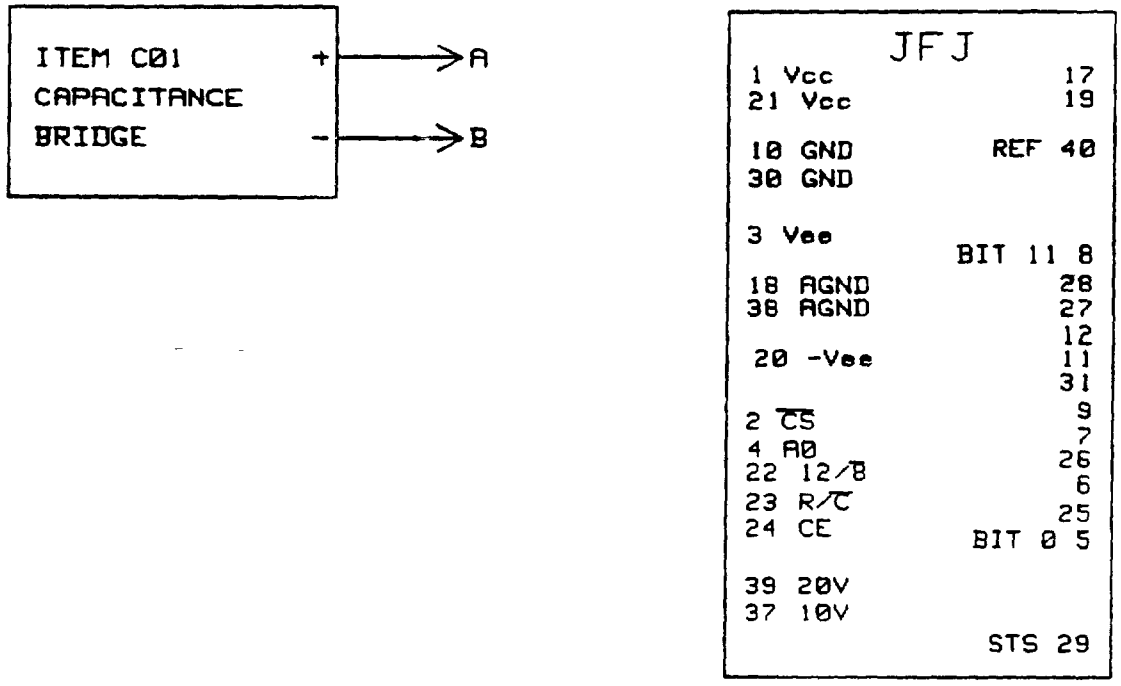


FIGURE 13. Power supply filter capacitance test circuit for table XII.

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TABLE XIII. Input capacitance test.

Test cond.	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	$\overline{CS}$	2	1	20	pF	1-3
M2	A0	4				
M3	R/$\overline{C}$	23				
M4	CE	24	1	20	pF	1-3

NOTES:

1. Input capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	M1-M4
Capacitance meter voltage	15 mV	± 1.5 mV	M1-M4
Capacitance measurement	NA	± 0.95 pF	M1-M4

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. Abbreviations used are defined in 6.1.

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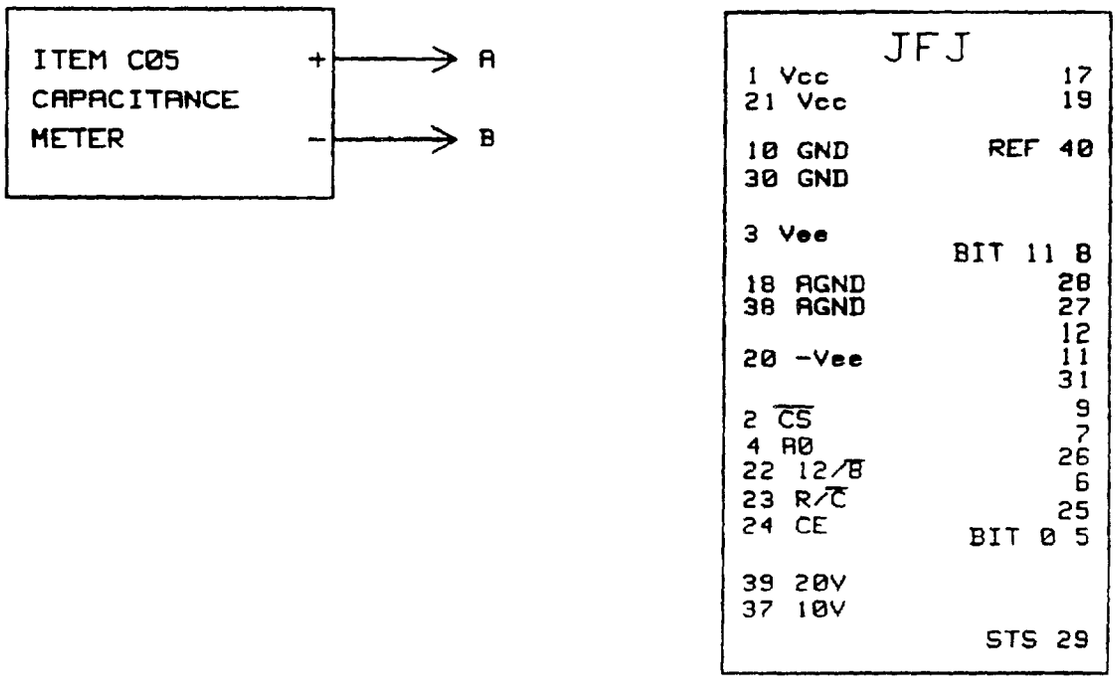


FIGURE 14. Digital input capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond.	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N1	5V (Vcc)	1,21	0.1	40.0	mA	1-3
N2	+15V (Vee)	3	0.1	7.0	mA	1-3
N3	-15V (-Vee)	20	-32.0	-0.1	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N1-N3
(Vee)	16.5 V	+ 0 mV,-165 mV	N1-N3
(-Vee)	-16.5 V	+ 165 mV,-0 mV	N1-N3
Power supply current limit (Vcc)	60 mA	NA	N1-N3
(Vee)	10.5 mA	NA	N1-N3
(-Vee)	-48 mA	NA	N1-N3
Pattern voltage high, (VH)	3.0 V	± 100 mV	N1-N3
low, (VL)	0.0 V	± 100 mV	N1-N3
Power supply current measurement	NA	± 1.99 mA	N1
	NA	± 0.35 mA	N2
	NA	± 1.59 mA	N3

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. 'A' shall correspond to the positive terminal and 'B' shall correspond to the negative terminal of the power supply. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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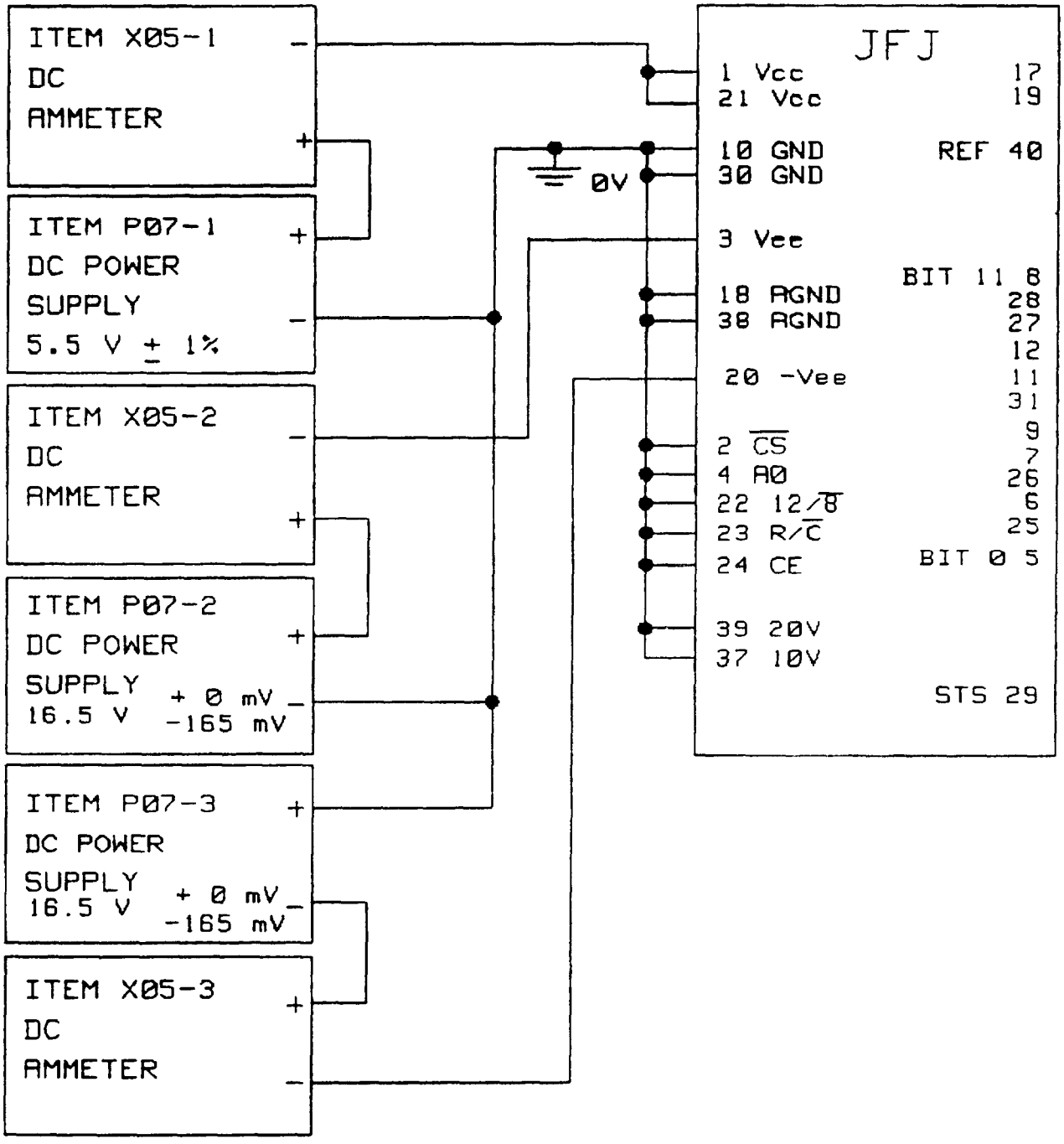


FIGURE 15. Power supply current test circuit for table XIV.

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TABLE XV. Unipolar accuracy test.

Test cond.	Parameter	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	Max	Units	Test fig.
P1	Zero (20 V)	39	Open	-0.0122	0.0171	V	16,17
P2	Zero (10 V)	37	Open	-0.0061	0.0085	V	16,17
P3	Midscale (10 V)	37	Open	4.9915	5.0061	V	16,17
P4	Midscale (20 V)	39	Open	9.9829	10.0122	V	16,17
P5	Full scale (10 V)	37	Open	9.9890	10.0037	V	16,17
P6	Full scale (20 V)	39	Open	19.9780	20.0073	V	16,17

NOTES:

1. Unipolar accuracy test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 50 mV	P1-P6
(Vee)	15.0 V	± 150 mV	P1-P6
(-Vee)	-15.0 V	± 150 mV	P1-P6
Power supply current limit (Vcc)	60 mA	NA	P1-P6
(Vee)	10.5 mA	NA	P1-P6
(-Vee)	-48 mA	NA	P1-P6

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage high, (VH) low, (VL)	 2.0 V 0.8 V	 ± 100 mV ± 40 mV	 P1-P6 P1-P6
Output voltage verification high, (Voh) low, (Vol)	 NA NA	 ± 105 mV ± 20 mV	 P1-P6 P1-P6

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. 'A' shall correspond to the positive terminal and 'B' shall correspond to the negative terminal of the power supply. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.4\ V$.

3. Abbreviations used are defined in 6.1.

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TABLE XVI. Bipolar accuracy test.

Test cond.	Parameter	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Test fig.
Q1	Zero (± 10 V)	39	19	-0.0171	0.0122	V	16,17
Q2	Zero (± 5 V)	37	19	-0.0085	0.0061	V	16,17
Q3	Neg. full scale (± 5 V)	37	19	-5.0061	-4.9915	V	16,17
Q4	Pos. full scale (± 5 V)	37	19	4.9890	5.0037	V	16,17
Q5	Neg. full scale (± 10 V)	39	19	-10.0122	-9.9829	V	16,17
Q6	Pos. full scale (± 10 V)	39	19	9.9780	10.0073	V	16,17

NOTES:

1. Bipolar accuracy test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 50 mV	Q1-Q6
(Vee)	15.0 V	± 150 mV	Q1-Q6
(-Vee)	-15.0 V	± 150 mV	Q1-Q6

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Power supply current limit (Vcc)	60 mA	NA	Q1-Q6
(Vee)	10.5 mA	NA	Q1-Q6
(-Vee)	-48 mA	NA	Q1-Q6
Pattern voltage high, (VH)	2.0 V	± 100 mV	Q1-Q6
low, (VL)	0.8 V	± 40 mV	Q1-Q6
Output voltage verification high, (Voh)	NA	± 105 mV	Q1-Q6
low, (Vol)	NA	± 20 mV	Q1-Q6

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. 'A' shall correspond to the positive terminal and 'B' shall correspond to the negative terminal of the power supply. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.4\ V$.

3. Abbreviations used are defined in 6.1.

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TABLE XVII. Linearity tests.

Test cond.	Parameter	25 ± 5°C Initial limits Min	Max	Units	Test fig.
R1-R4095	Integral linearity (all codes)	-0.5	0.5	LSB	16-18

NOTES:

1. Linearity test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 50 mV	R1-R4095
(Vee)	15.0 V	± 150 mV	R1-R4095
(-Vee)	-15.0 V	± 150 mV	R1-R4095
Power supply current limit (Vcc)	60 mA	NA	R1-R4095
(Vee)	10.5 mA	NA	R1-R4095
(-Vee)	-48 mA	NA	R1-R4095
Pattern voltage high, (VH)	2.0 V	± 100 mV	R1-R4095
low, (VL)	0.8 V	± 40 mV	R1-R4095
Output voltage verification high, (Voh)	NA	± 105 mV	R1-R4095
low, (Vol)	NA	± 20 mV	R1-R4095

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress table. 'A' shall correspond to the positive terminal and 'B' shall correspond to the negative terminal of an external power supply. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.4\ V$.

3. Abbreviations used are defined in 6.1.

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TABLE XVIII. Differential linearity test.

Test cond.	Parameter	25 ± 5°C Initial limits		Units	Test fig.
		Min	Max		
S1-S4095	Differential linearity (all codes)	-0.75	0.75	LSB	16,17, 19

NOTES:

1. This test uses the data obtained in the integral linearity test. The difference of two consecutive points shall meet the requirements in table XVIII.

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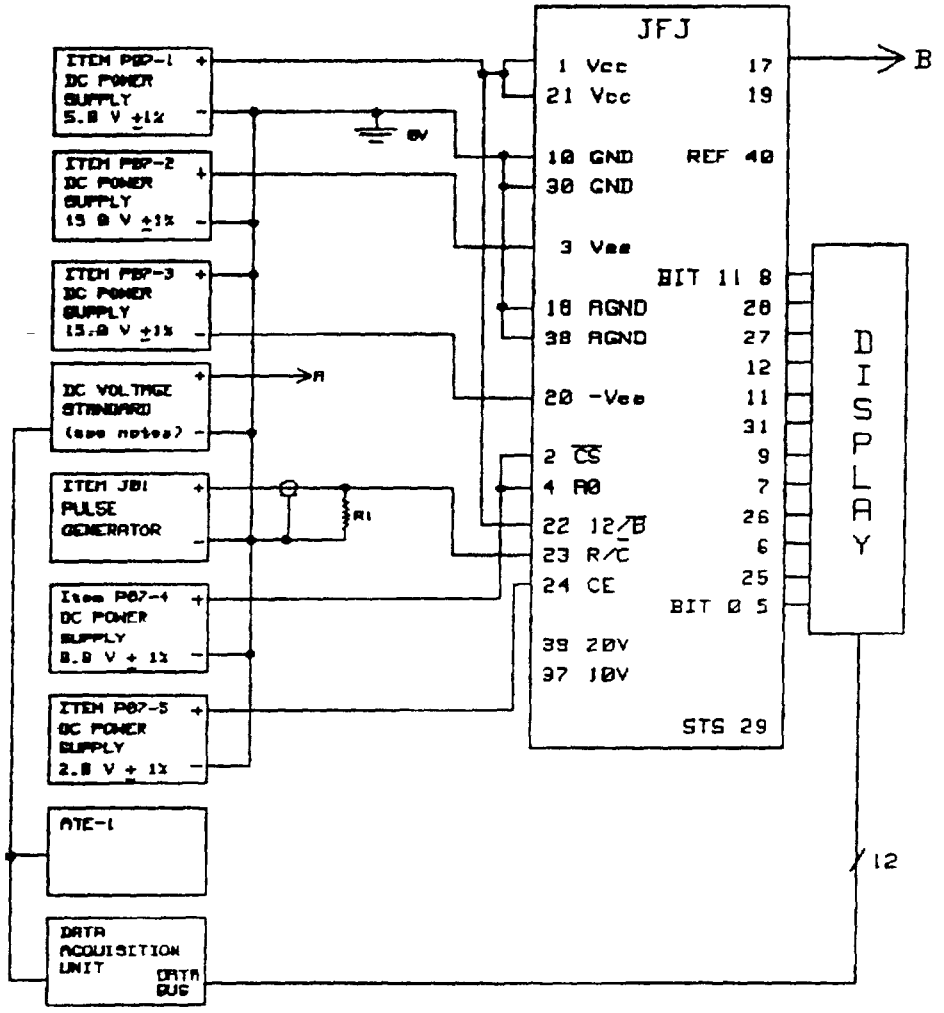


FIGURE 16. Circuit for unipolar and bipolar accuracy and all codes linearity tests for tables XV, XVI, XVII, and XVIII.

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NOTES:

1. R1=50 Ohms ±1 percent, 1W

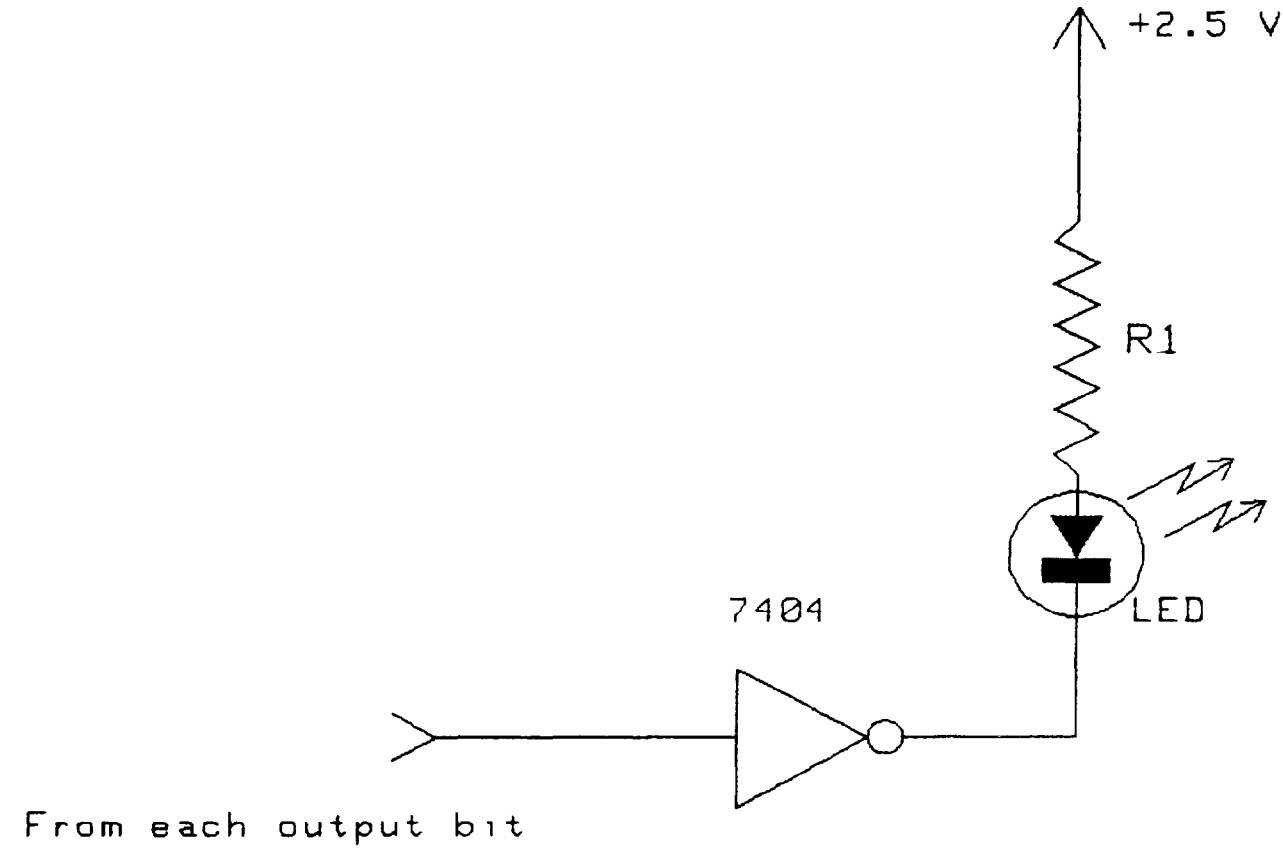
2. Item J01 settings:
   Inverted pulse
   Vmax = 3.0 ± 0.1 V
   Vmin = 0.0 ± 0.1 V
   PW = 1 ± 0.1 uS
   tTLH = 20 ± 2 nS
   tTHL = 20 ± 2 nS
   PRP = 50 ± 5 uS

3. See figure 3 for waveforms.

4. DISPLAY shall be connected to a data analyzer which is controllable through the IEEE 488 parallel bus.

5. The dc voltage standard shall be programmable via the IEEE 488 bus.

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NOTE:

1. R1 = 2 K Ohms

FIGURE 17. Display circuit.

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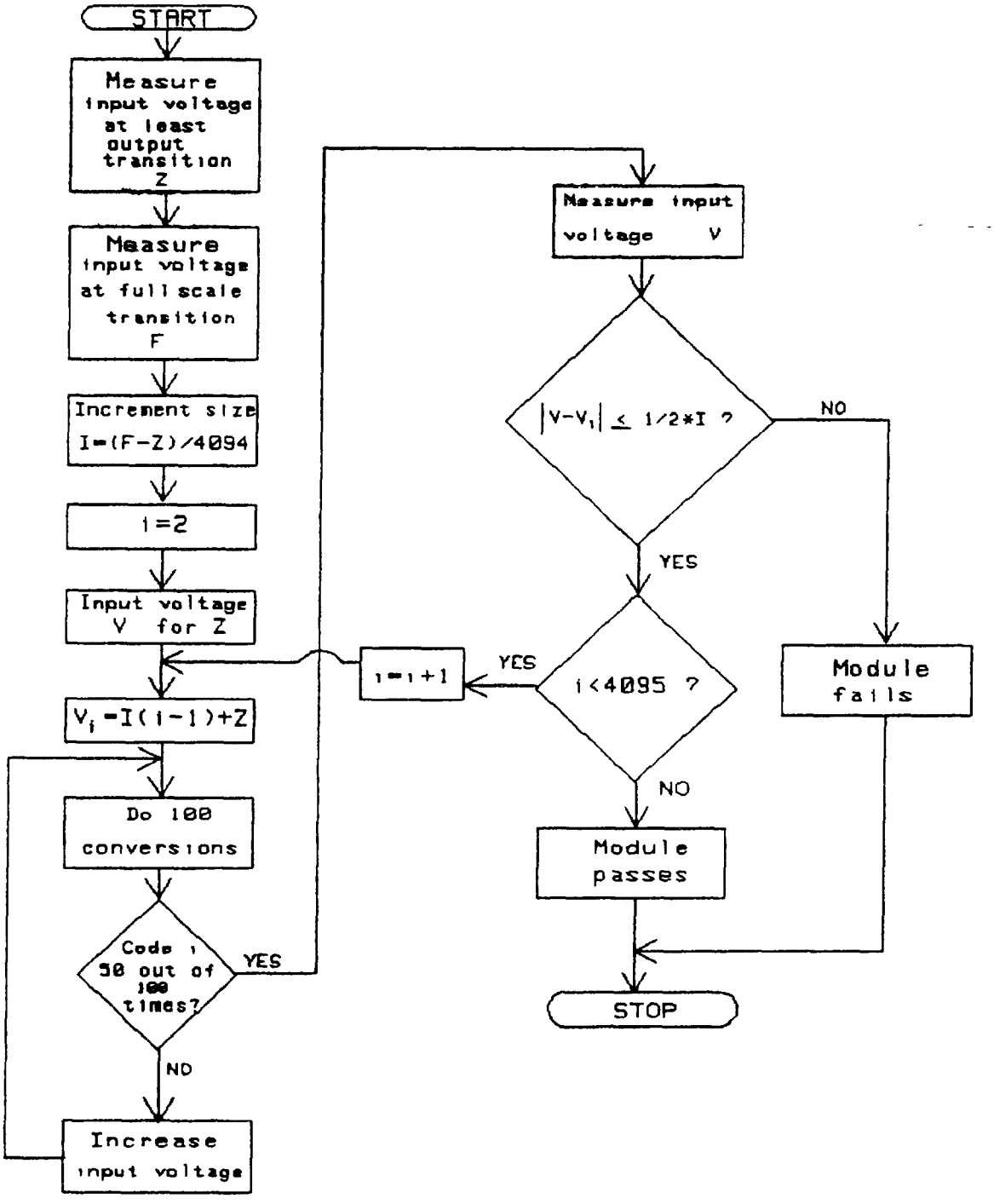


FIGURE 18. Integral linearity test algorithm.

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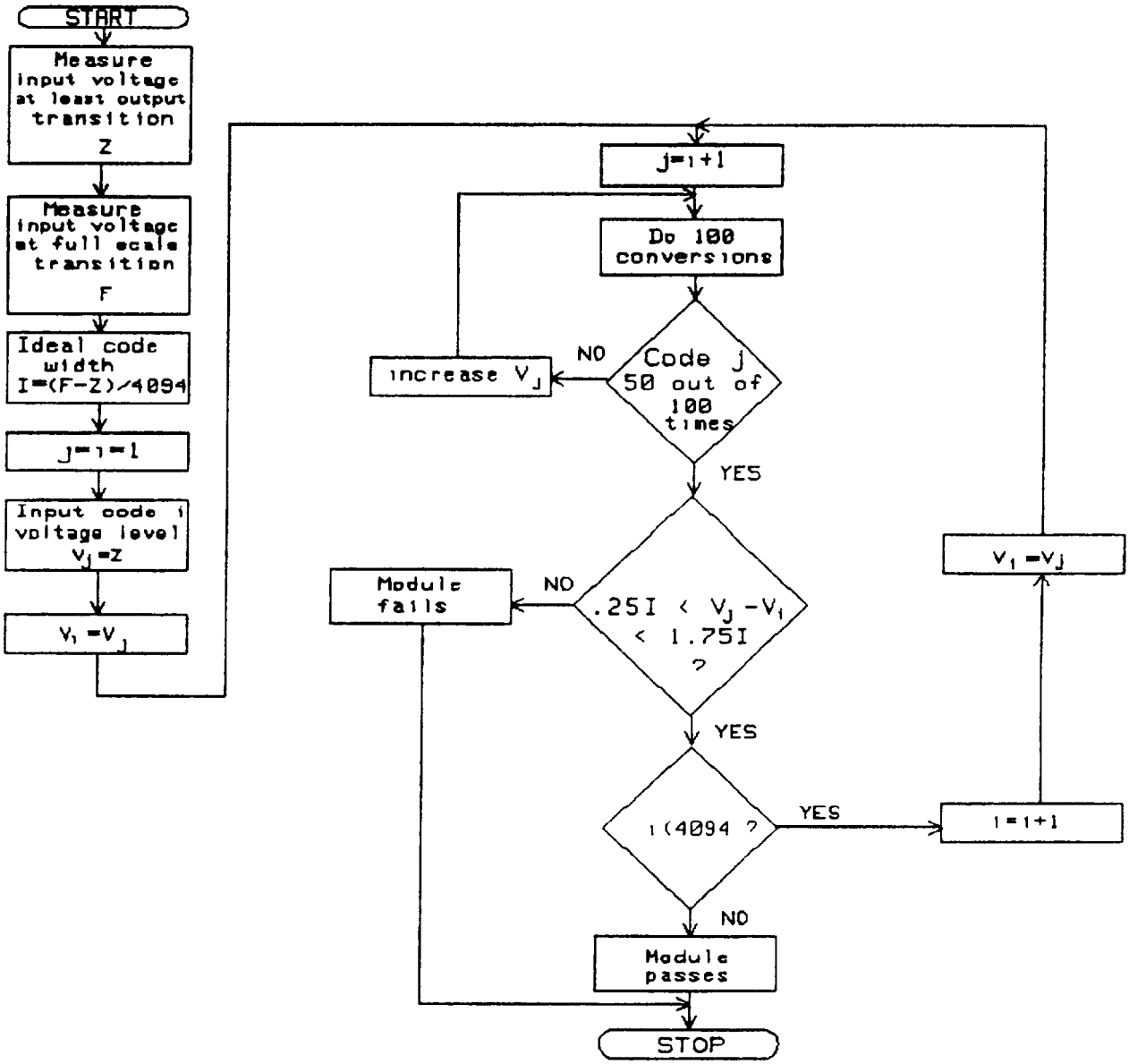


FIGURE 19. Differential linearity test algorithm.

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TABLE XIX. 8-bit conversion test.

Test cond.	Parameter	Connect A to pin	Connect B to pin	Output word 1110 9 8 7 6 5 4 3 2 1 0	Test fig.
T	8-bit conversion	Open 4	4 Open	H H H H H H H H X X X X X X X X L L L L H H H H	20

NOTES:

1. 8-bit conversion test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 50 mV	T
(Vee)	15.0 V	± 150 mV	T
(-Vee)	-15.0 V	± 150 mV	T
Power supply current limit (Vcc)	60 mA	NA	T
(Vee)	10.5 mA	NA	T
(-Vee)	-48 mA	NA	T
Pattern voltage high, (VH)	3.0 V	± 120 mV	T
low, (VL)	0.0 V	± 20 mV	T
Output voltage verification high, (Voh)	NA	± 105 mV	T
low, (Vol)	NA	± 20 mV	T

2. 'X' indicates a 'don't care' state.

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NOTES: - Continued.

3. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. 'A' shall correspond to the positive terminal and 'B' shall correspond to the negative terminal of the power supply. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Before performing this test, trigger item J01. Item J01 shall not require triggering thereafter.

(c) The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.4\ V$.

4. Abbreviations used are defined in 6.1.

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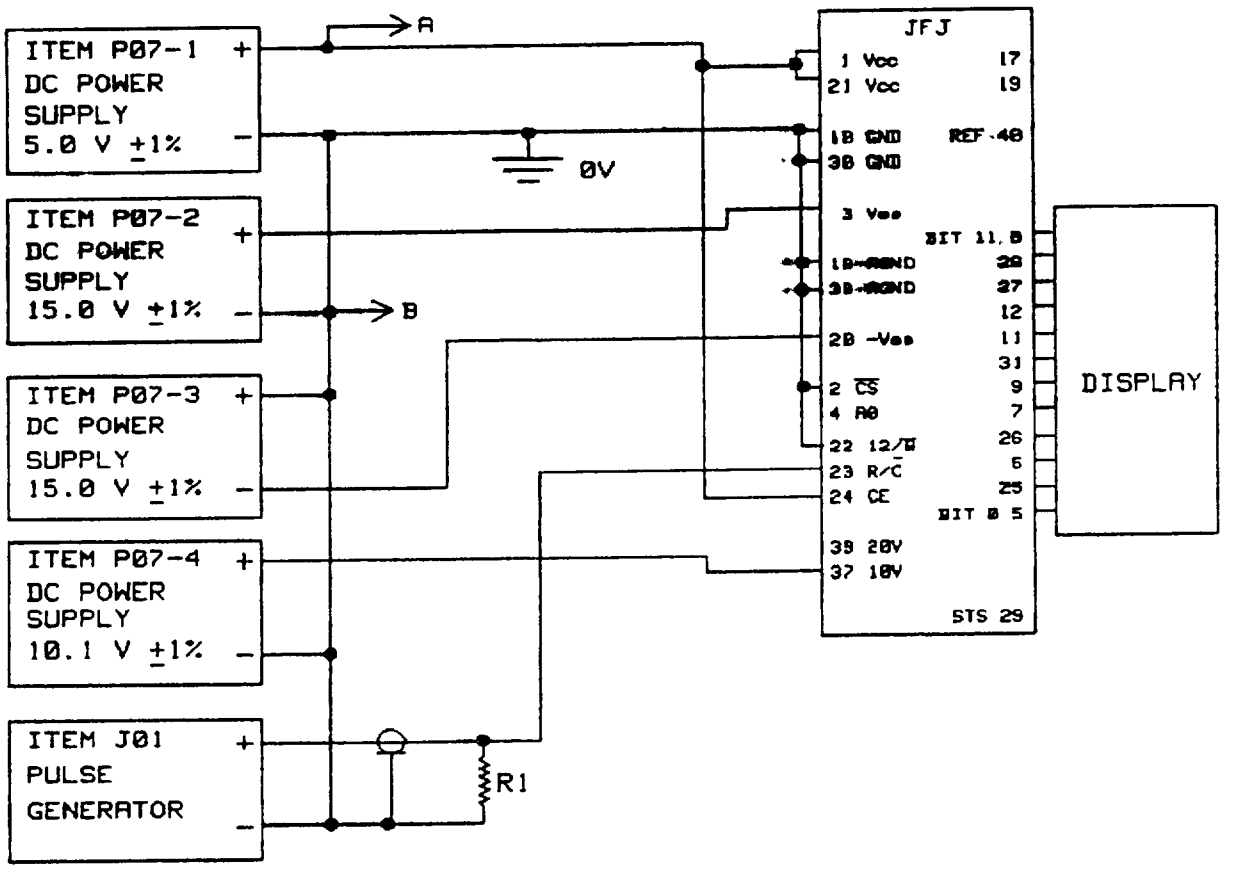


FIGURE 20. 8-bit conversion test circuit for table XIX.

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NOTES:

1. R1 = 50 Ohms ± 1 percent, 1 W

2. Item J01 settings:
   Inverted pulse
   Vmax = 3.0 ± 0.1 V
   Vmin = 0.0 ± 0.1 V
   PW = 1 ± 0.1 uS
   tTLH = 20 ± 2 nS
   tTHL = 20 ± 2 nS
   Single shot

3. See figure 3 for waveforms.

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TABLE XX. Conversion cycle timing tests.

Test cond.	Parameter	Symbol	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	Max	Units	Test fig.
U1	STS to CE delay	tDSC	1	24	29	0	300	nS	21,22
U2	Conv. time (8-bit)	tc	10	Open	29	10	24	uS	21,22
U3	Conv. time (12-bit)	tc	1	Open	29	15	35	uS	21,22

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TABLE XXI. Read cycle timing tests.

Test cond.	Parameter	Symbol	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Test fig.
V1	Data out, CE low	tHL	1	24	8	25	150	nS	21,22
V2	Access time	tDD	1	24	8	0	250	nS	21,22

NOTES:

1. Conversion and read cycle test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 50 mV	U1-U3, V1, V2
(Vee)	15.0 V	± 150 mV	U1-U3, V1, V2
(-Vee)	-15.0 V	± 150 mV	U1-U3, V1, V2
Power supply current limit (Vcc)	60 mA	NA	U1-U3, V1, V2
(Vee)	10.5 mA	NA	U1-U3, V1, V2
(-Vee)	-48 mA	NA	U1-U3, V1, V2
Input voltage high, (VH)	2.4 V	± 120 mV	U1-U3, V1, V2
low, (VL)	0.4 V	± 20 mV	U1-U3, V1, V2
Output voltage verification high, (Voh)	NA	± 105 mV	U1-U3, V1, V2
low, (Vol)	NA	± 20 mV	U1-U3, V1, V2

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NOTES: - Continued.

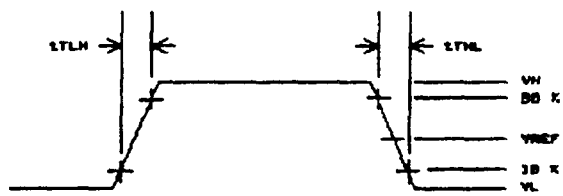
2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. 'A' shall correspond to the positive terminal and 'B' shall correspond to the negative terminal of the power supply. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.4\ V$.

3. Input/Output (I/O) waveforms are as specified on figure 22. All of the input waveforms shall have the following characteristics:



Where:

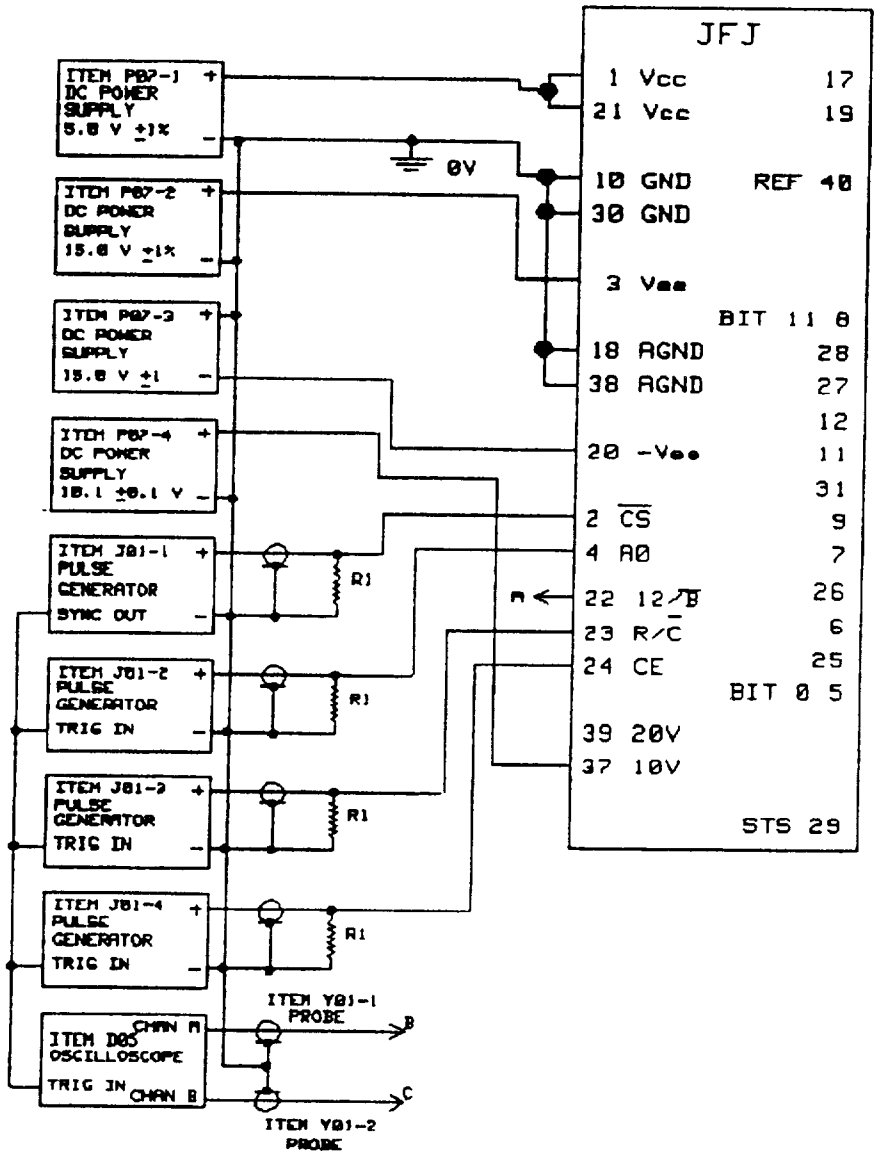
(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 20 nS.

(c) Vref = 1.5 V.

4. Abbreviations used are defined in 6.1.

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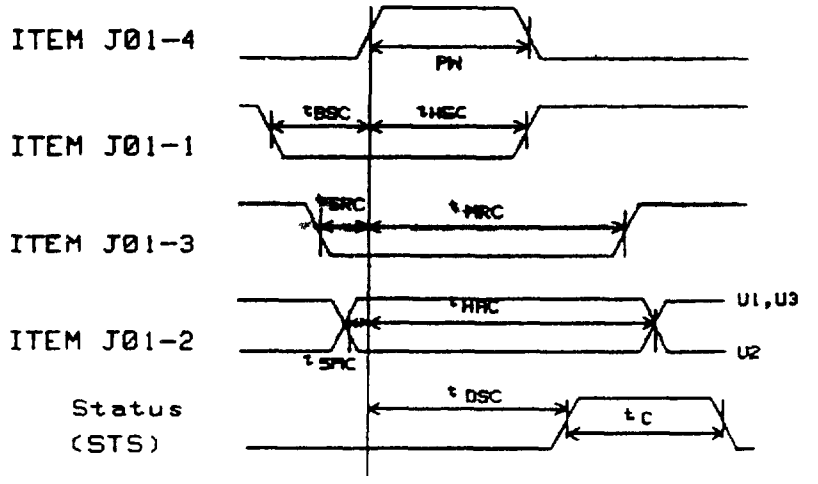
NOTES:

1. R1= 50 Ohms ± 1 percent, 1 W.
2. See figure 22 for pulse generator set-up.

FIGURE 21. Timing test circuit for tables XX and XXI.

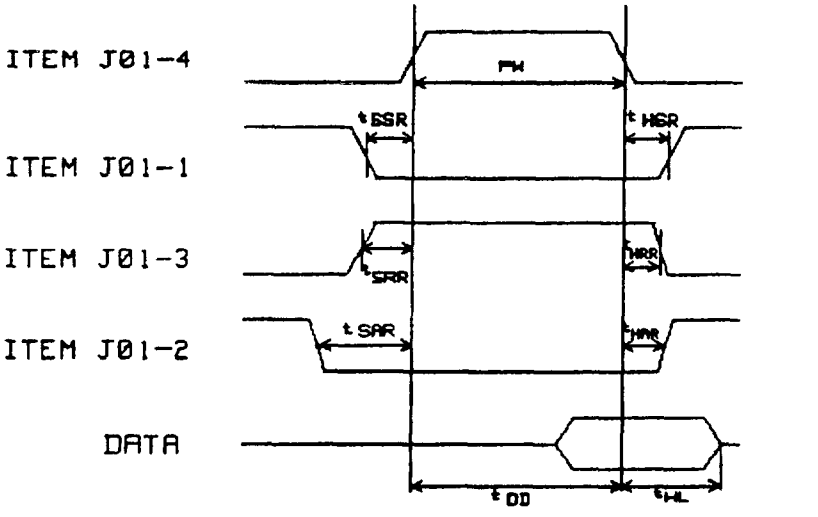
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Convert cycle (test conditions U1-U3)



PW ≥ 300 nS
tSSC ≥ 200 nS
tHSC ≥ 200 nS
tSRC ≥ 250 nS
tHRC ≥ 200 nS
tSAC = 0 nS
tHAC ≥ 300 nS

Read cycle (test conditions V1 and V2)



tSSR ≥ 150 nS
tHSR ≥ 50 nS
tSRR ≥ 0 nS
tHRR ≥ 0 nS
tSAR ≥ 150 nS
tHAR ≥ 50 nS

NOTE:
1. The repetition rate, PRP, for each pulse generator is 50 ± 5 uS.

FIGURE 22. Timing diagram set-up for tables XX and XXI.

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TABLE XXII. Analog input impedance test.

Test cond.	Parameter	S1 position	Connect A to pin	25 ± 5°C Initial limit Min	Max	Units	Test fig.
W1	20 V input	1	39	1.85	3.15	V	23
W2	10 V input	2	37	1.85	3.15	V	23

NOTES:

1. Analog input impedance test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply (Vcc)	5.0 mV	± 50 mV	W1, W2
(Vee)	15.0 mV	± 150 mV	W1, W2
(-Vee)	-15.0 mV	± 150 mV	W1, W2
Power supply current limit (Vcc)	60 mA	NA	W1, W2
(Vee)	10.5 mA	NA	W1, W2
(-Vee)	-48 mA	NA	W1, W2
Input voltage measurement	NA	± 10 mV	W1, W2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. 'A' shall correspond to the positive terminal and 'B' shall correspond to the negative terminal of an external power supply. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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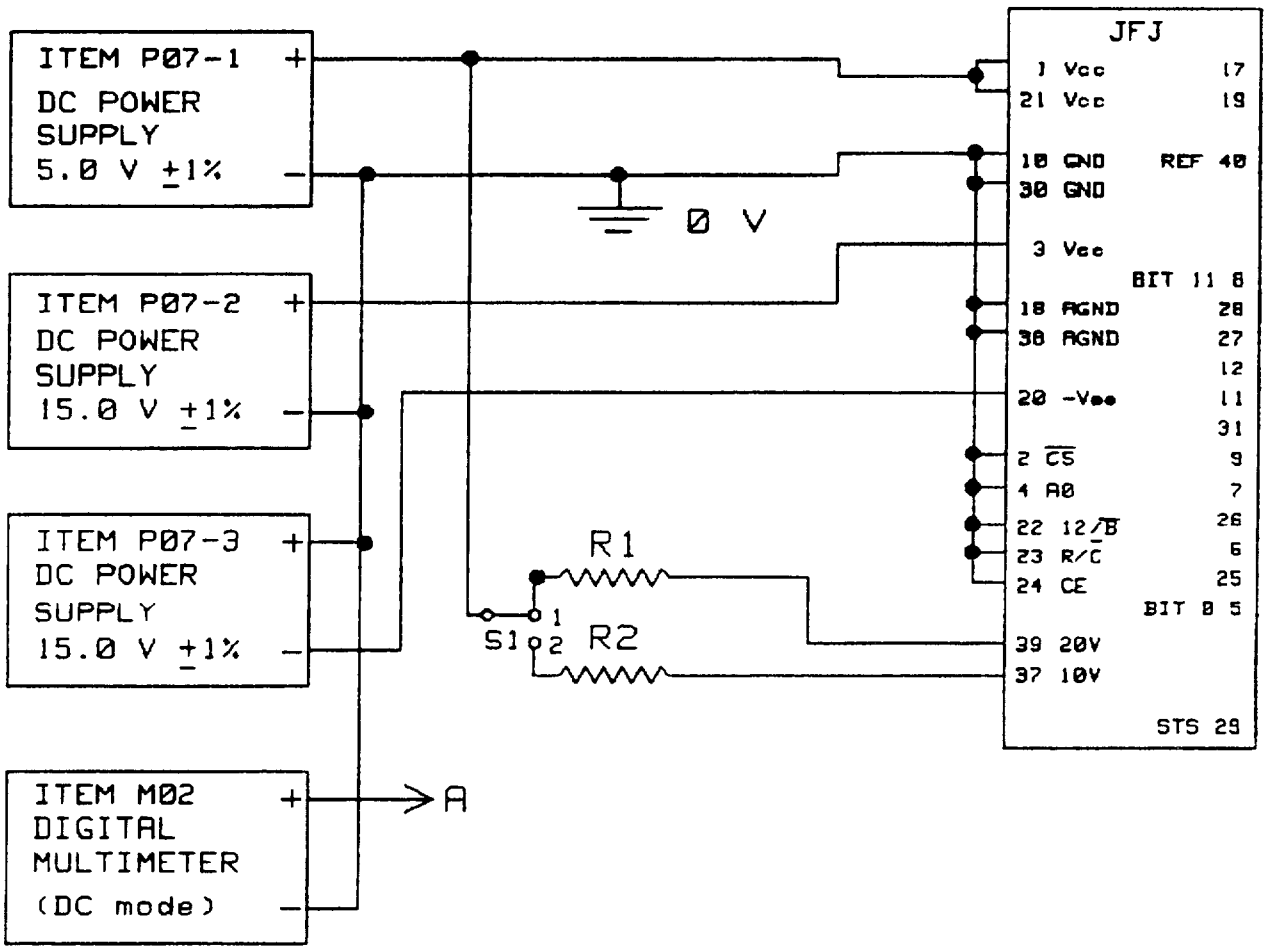
NOTES: - Continued

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

3. The input impedance $Z = \frac{WR}{5-W}$.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. R1 = 10.0 K Ohm ± 1 percent, 1/8 W.
2. R2 = 5.00 K Ohm ± 1 percent, 1/4 W.
3. S1 = SPDT switch.

FIGURE 23. Input impedance test circuit for table XXII.

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TABLE XXIII. Reference voltage, loaded, test.

Test cond.	Parameter	25 ± 5°C Initial limit Min	Max	Units	Test fig.
X	Reference voltage, loaded	9.95	10.05	V	24

NOTES:

1. Reference voltage, loaded, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	X
(Vee)	13.5 V	± 135 mV	X
(-Vee)	-13.5 V	± 135 mV	X
Power supply current limit (Vcc)	60 mA	NA	X
(Vee)	10.5 mA	NA	X
(-Vee)	-48 mA	NA	X
Output voltage measurement	NA	± 10 mV	X

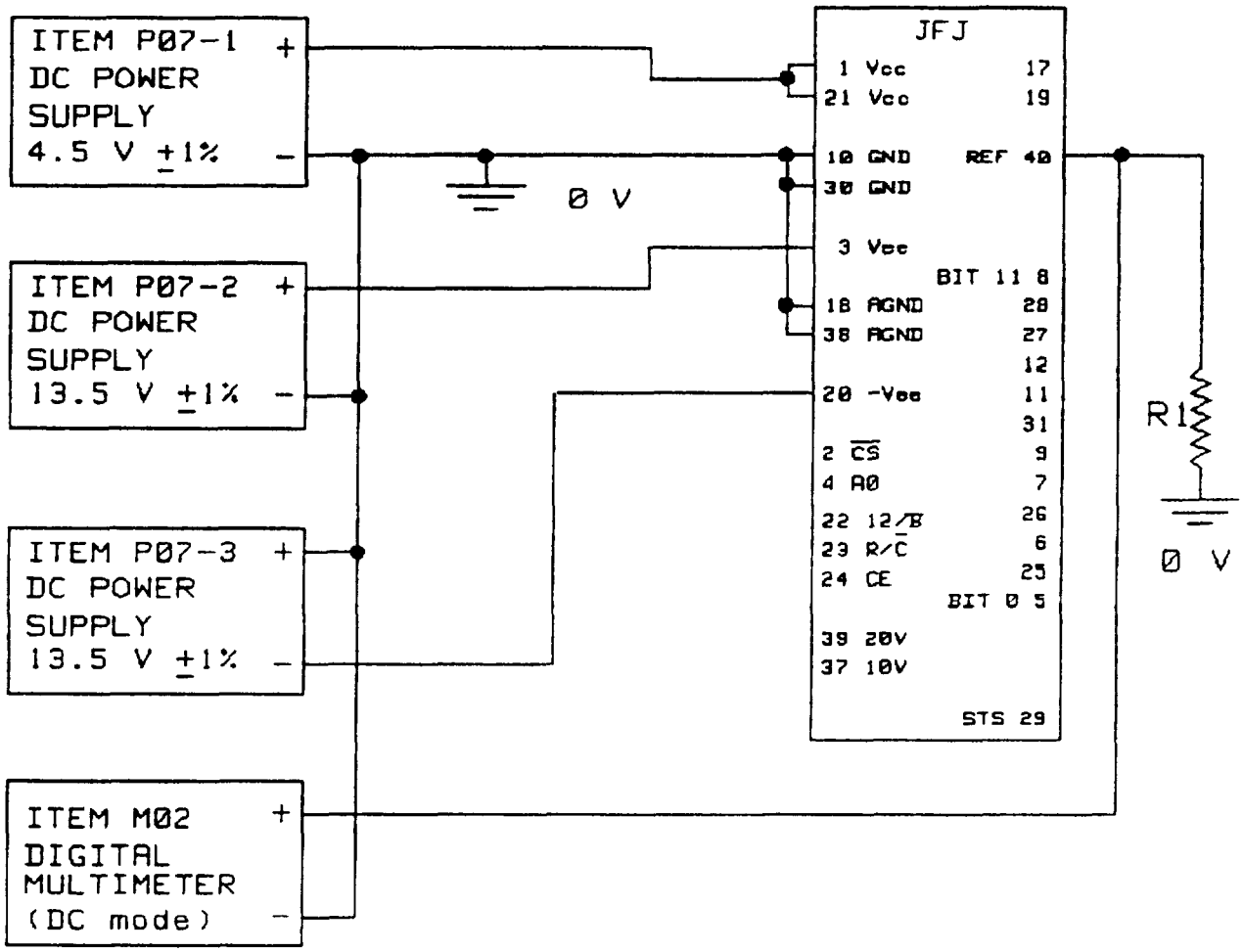
2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. 'A' shall correspond to the positive terminal and 'B' shall correspond to the negative terminal of the power supply. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

3. Abbreviations used are defined in 6.1.

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NOTE:

1. R1 = 10.0 K Ohm ± 5 percent, 1/8 W.

FIGURE 24. Reference voltage, loaded test circuit for table XXIII.

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TABLE XXIV. Power supply transient amplitude test.

Test cond.	Parameter	Connect A to pin	25 ± 5°C Initial limit Min	Max	Units	Test fig.
Y1	PSTA (5 V)	1	0	400	mVp	25-27
Y2	PSTA (+15 V)	3	0	700	mVp	25-27
Y3	PSTA (-15 V)	20	0	700	mVp	25-27

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.25 V	Y1-Y3
(Vee)	15.0 V	± 0.75 V	Y1-Y3
(-Vee)	-15.0 V	± 0.75 V	Y1-Y3
Power supply current limit (Vcc)	60 mA	NA	Y1-Y3
(Vee)	10.5 mA	NA	Y1-Y3
(-Vee)	-48 mA	NA	Y1-Y3
Pattern voltage high, (VH)	3.0 V	± 100 mV	Y1-Y3
low, (VL)	0.0 V	± 100 mV	Y1-Y3
Power supply transient amplitude measurement	NA	± 20 mV	Y1
		± 35 mV	Y2, Y3

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. 'A' shall correspond to the positive terminal and 'B' shall correspond to the negative terminal of the power supply. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use probe 'B' to verify the pattern voltage and pattern repetition rate indicated on figure 26.

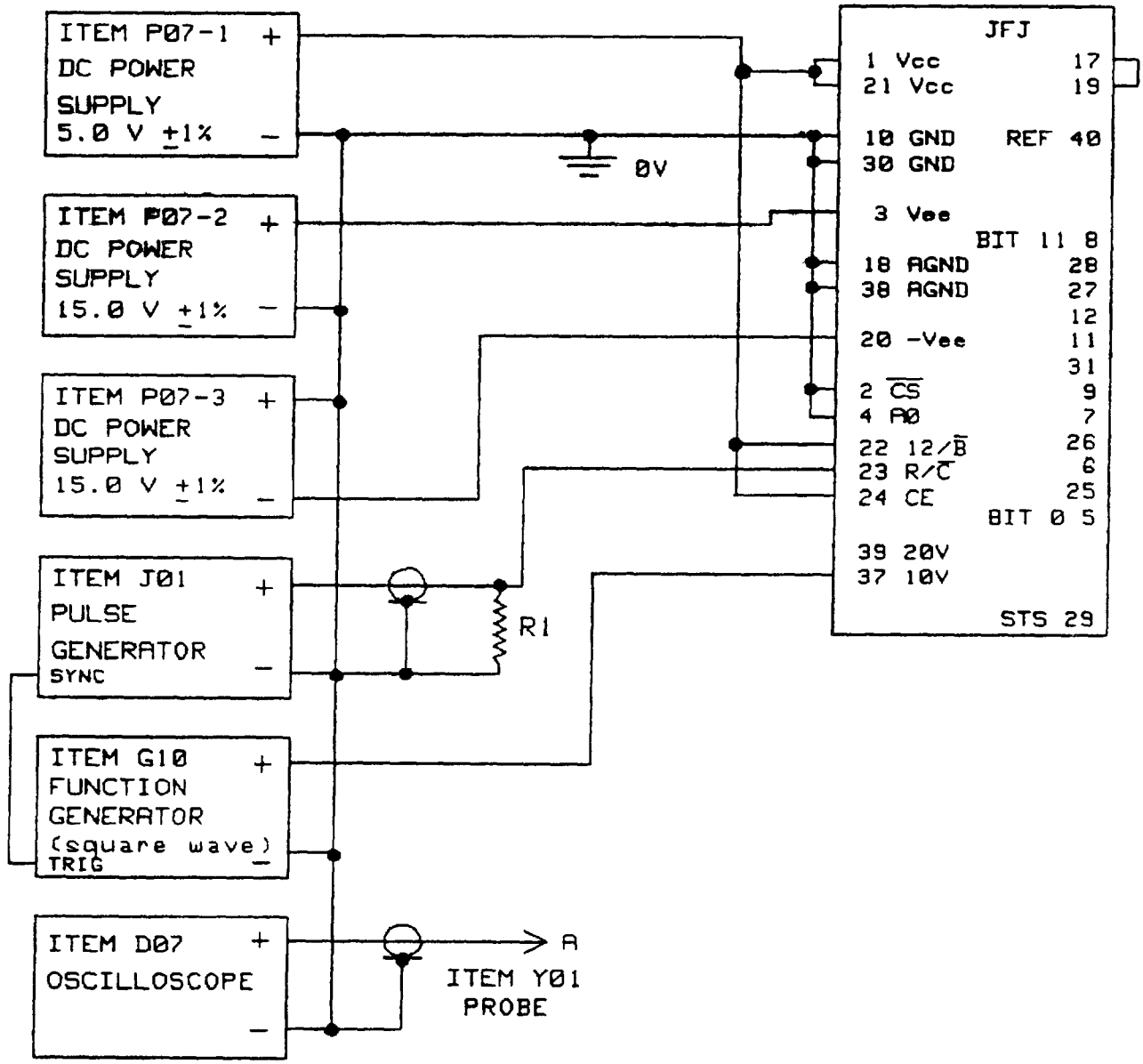
(c) All Vcc pins shall be shorted together and the scope point, probe 'A', connected to Vcc at the point of shorting.

(d) Using the oscilloscope, D07, set at a maximum strobe rate of one-tenth of the cycle time per division, visually inspect each pattern row for the most negative spike as indicated on figure 27.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. Abbreviations used are defined in 6.1.

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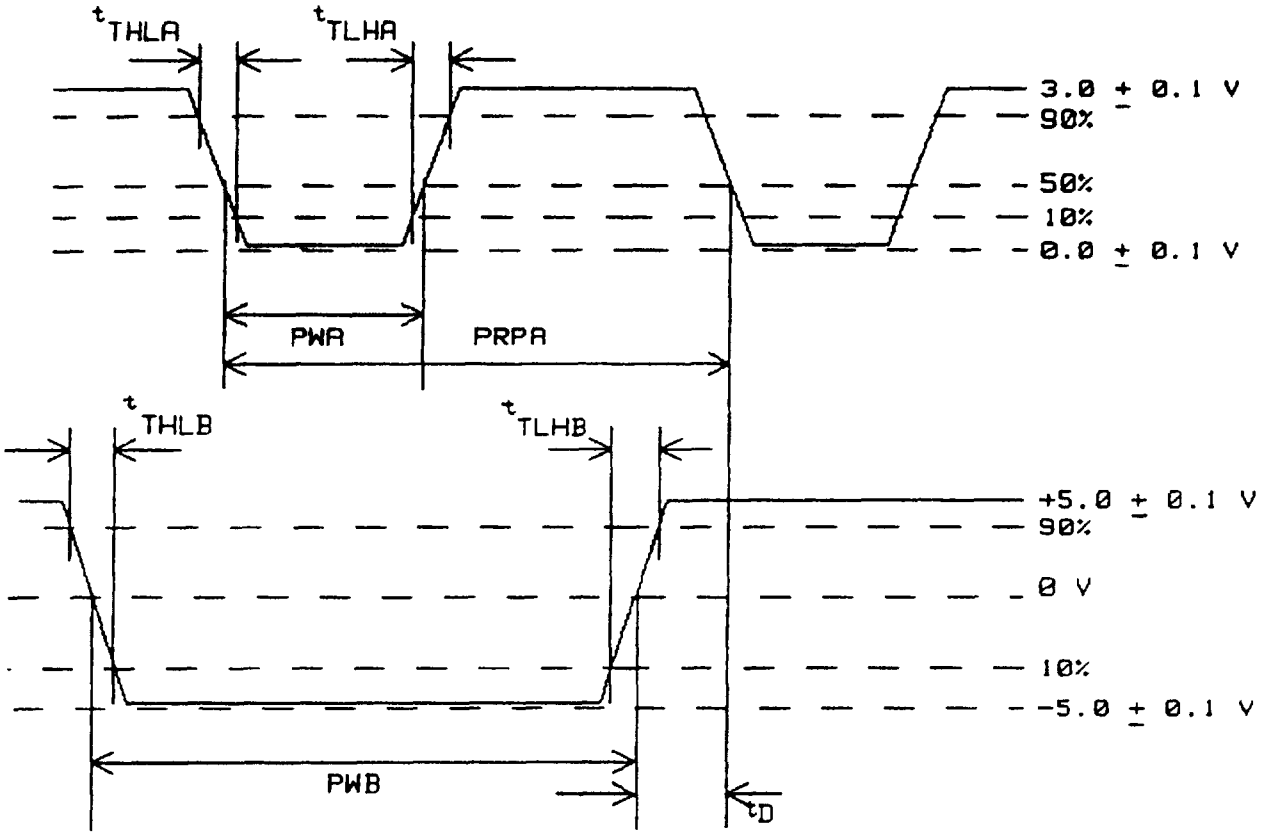


NOTES:

1. R1 = 50 Ohms ± 1 percent, 1 W;
2. See figure 26 for waveforms.

FIGURE 25. Power supply transient amplitude test circuit for table XXIV.

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NOTES:

1. Item J01:
   PWA = 1.0 ± 0.1 uS
   PRPA = 50 ± 5 uS
   tTLHA = 20 ± 2 nS
   tTHLA = 20 ± 2 nS

2. Item G10:
   PWB = 50 ± 5 uS
   PRPB = 100 ± 10 uS
   tTLHB = 40 ± 2 nS
   tTHLB = 40 ± 2 nS

3. tD = 50 ± 5 nS

FIGURE: 26. Timing circuit set-up for table XXIV.

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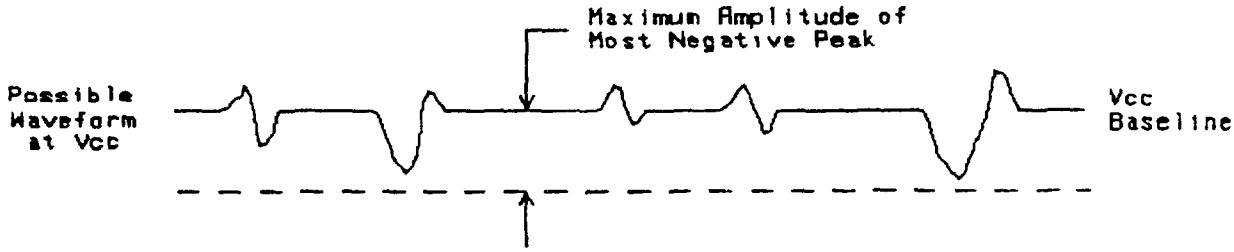


FIGURE 27. Power supply transient amplitude waveform for table XXIV.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-M-28787/419	2 DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE
Modules, Standard Electronic, Converter, A/D Module, Key Code JFJ

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed)

5 REASON FOR RECOMMENDATION

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a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
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8. PREPARING ACTIVITY

a NAME Technical Point of Contact (TPOC): Mr. Robert F. Dilger (Code 602A) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS	b TELEPHONE (Include Area Code) (1) Commercial TPOC: 812-854-1528 (2) AUTOVON 88-482-1528
c ADDRESS (Include Zip Code) Commanding Officer Naval Weapons Support Center ATTN: Robert F. Dilger, Bldg. 2940W, Code 602A Crane, Indiana 47522-5060	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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INCH-POUND

MIL-M-28787/42B(SH)
AMENDMENT 2
19 March 1992

SUPERSEDING
AMENDMENT 1
20 September 1983

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

COMPARATOR, MAGNITUDE, DIGITAL, KEY CODE JBK

This amendment forms a part of Military Specification MIL-M-28787/42B(EC), dated 14 May 1982, and is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

PAGE 4

* TABLE I, Test conditions G1-G7, Test type column Delete "X" and substitute "S"

* TABLE I, Test condition column Delete "P4-P39" and substitute "P4-P19, P24-P27, P32-P39"

* TABLE I, After test conditions P4-P19, P24-P27, P32-P39 insert the following "|P20-P23, P28-P31| Propagation delay time (t_{PLH}) C_L = 50pF |IX|S|10|200|10|200|10|220|10|220|ns|"

PAGE 6

* 4 4.2 Delete and substitute the following

"4.4.2 Order of testing. Test conditions A through X shall be performed in any order with the following exceptions. Test conditions A, B, C, and G are stress tests and shall be performed only at 25 ± 5°C If test conditions A, B, C, or G are part of a sequence of tests, the order of testing shall be as follows A, B, C, and G where applicable, followed by the remaining tests in the sequence "

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* 4 5 1, fifth line: Delete "W" and substitute "F, and H through W"

* 4 5 3, third line Delete "W" and substitute "F, and H through W"

* 4.5 3, sixth line Delete "X" and substitute "F, and H through W"

PAGE 15

FIGURE 5 Delete and substitute new figure 5 as printed on page 3 of this amendment

PAGE 32

TABLE VII, footnote 2, first line Delete "voltmeter V03" and substitute "item M02"

PAGE 33

FIGURE 9 Delete and substitute new figure 9 as printed on page 4 of this amendment

PAGE 41

* TABLE IX, Test conditions P20-P23, Max column Delete "150" and substitute "200"

* TABLE IX, Test conditions P24–P27, Max column Insert "150"

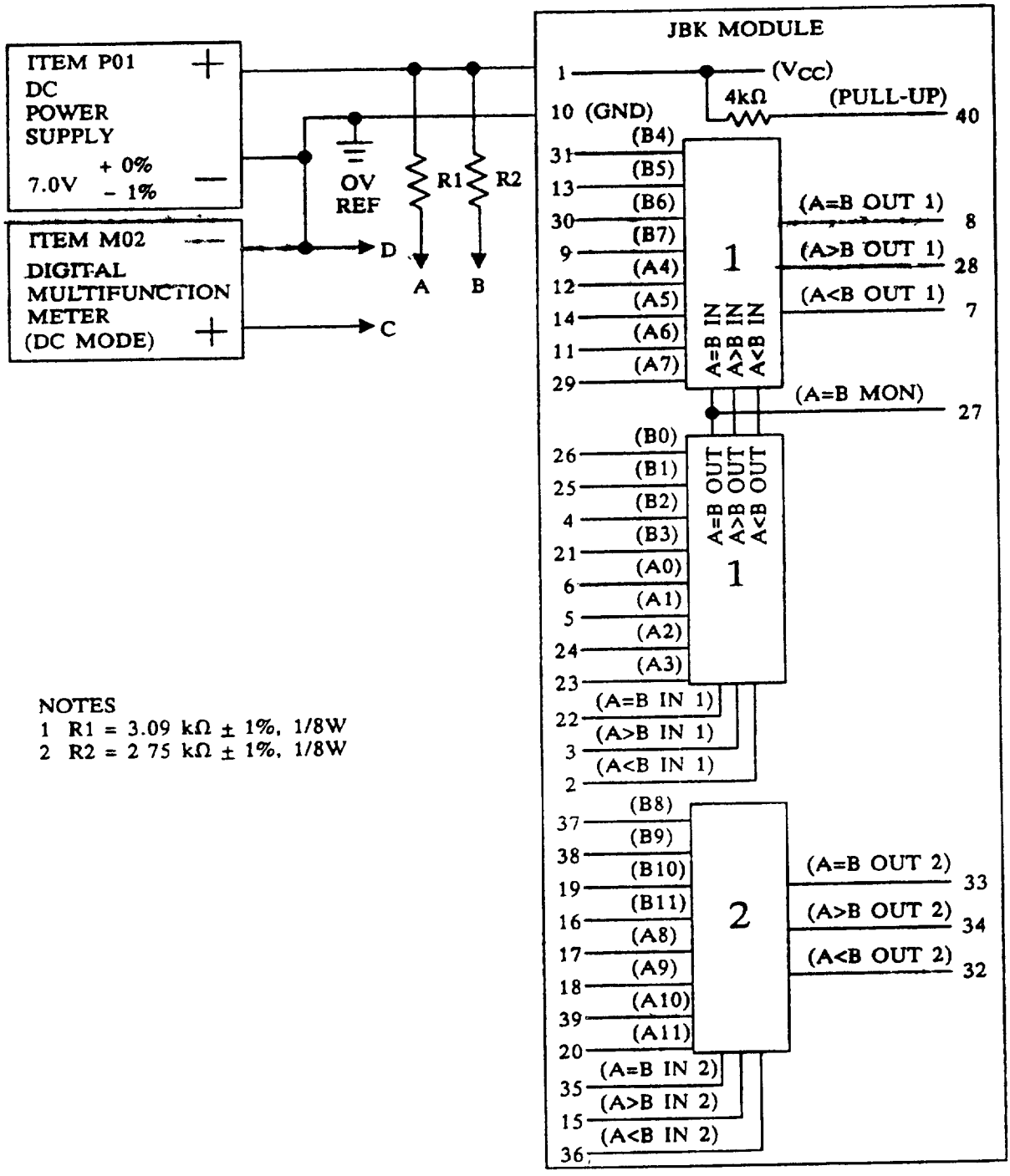
* TABLE IX, Test conditions P28-P30, Max column Delete "150" and substitute "200"

PAGE 42

* TABLE IX, Test condition P31, Max column Delete "150" and substitute "200"

* TABLE IX, Test conditions P32–P39, Max column Insert "150"

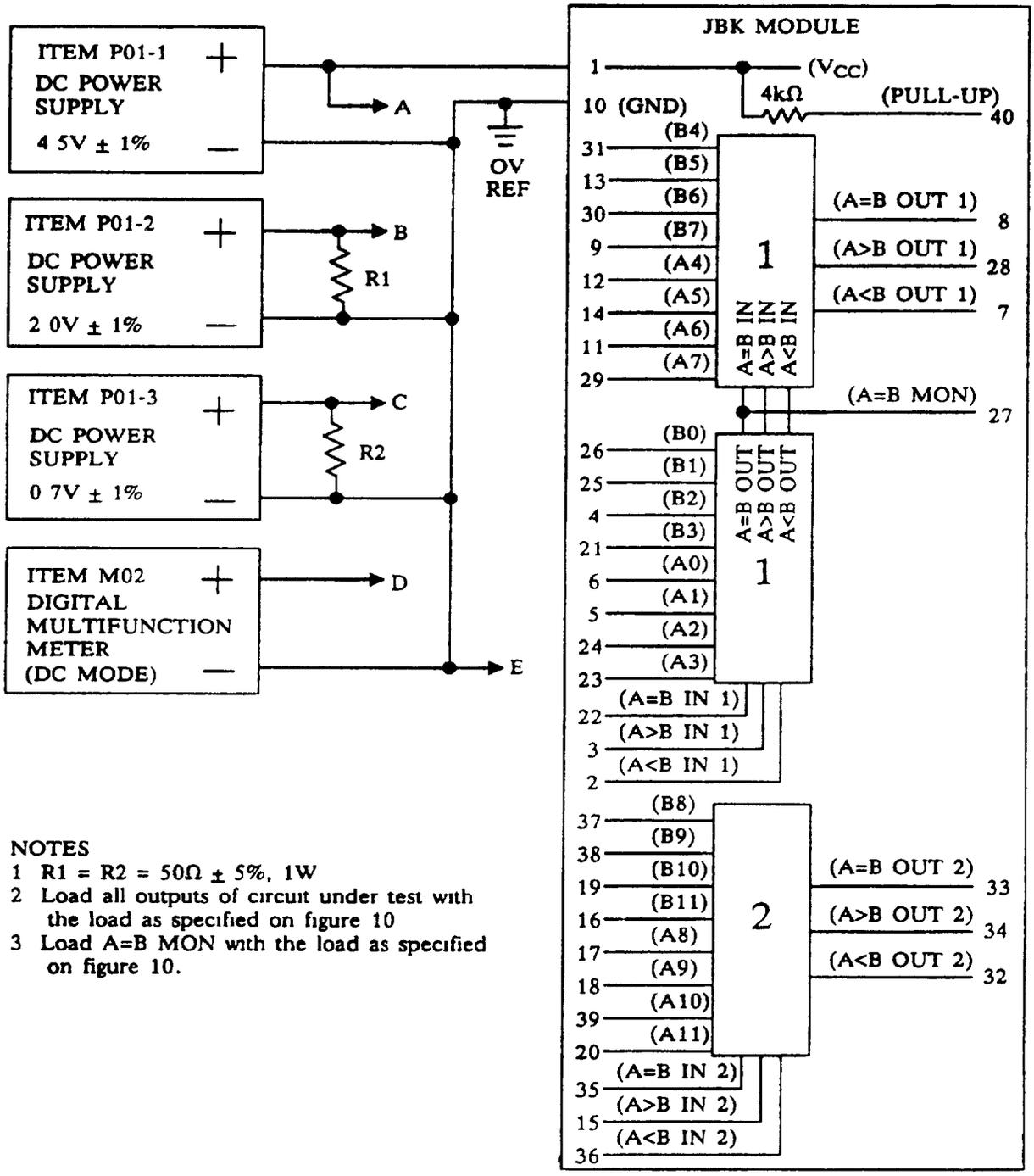
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NOTES
1 R1 = 3.09 kΩ ± 1%, 1/8W
2 R2 = 2 75 kΩ ± 1%, 1/8W

FIGURE 5 V_{OUT} (low-loaded) test circuit for table III

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NOTES
1 R1 = R2 = 50Ω ± 5%, 1W
2 Load all outputs of circuit under test with the load as specified on figure 10
3 Load A=B MON with the load as specified on figure 10.

FIGURE 9 Functional compare and output voltage (high-level and low-level) test circuit for table VII

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